



Introdução Axway API Manager

Workshop v1.1

Fernando Braga

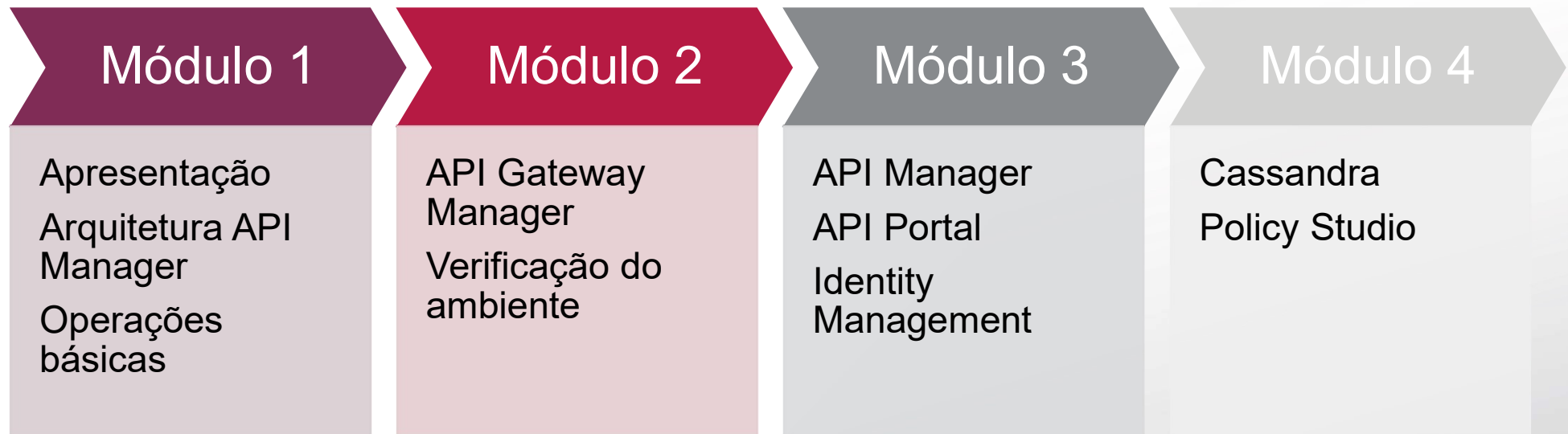
31/Ago/2022

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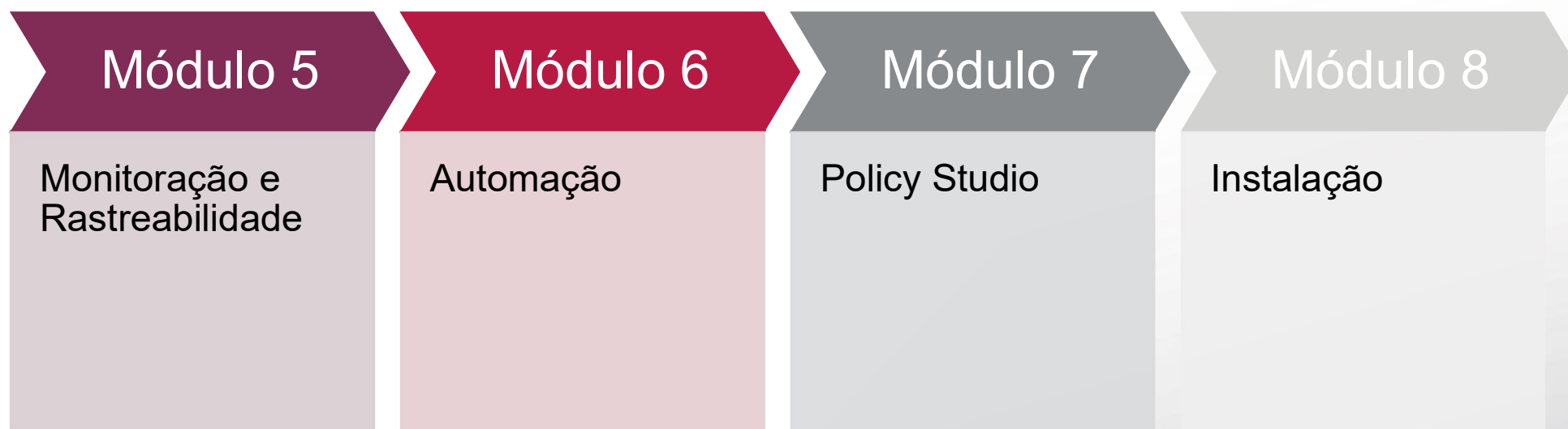
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Agenda



Agenda





API Gateway Manager

API Gateway Manager

Agenda

Introdução

Dashboard

Tráfego

Logs and Events

Outras Funcionalidades

Verificação Padrão

Integration with 3rd Party

Other Cases

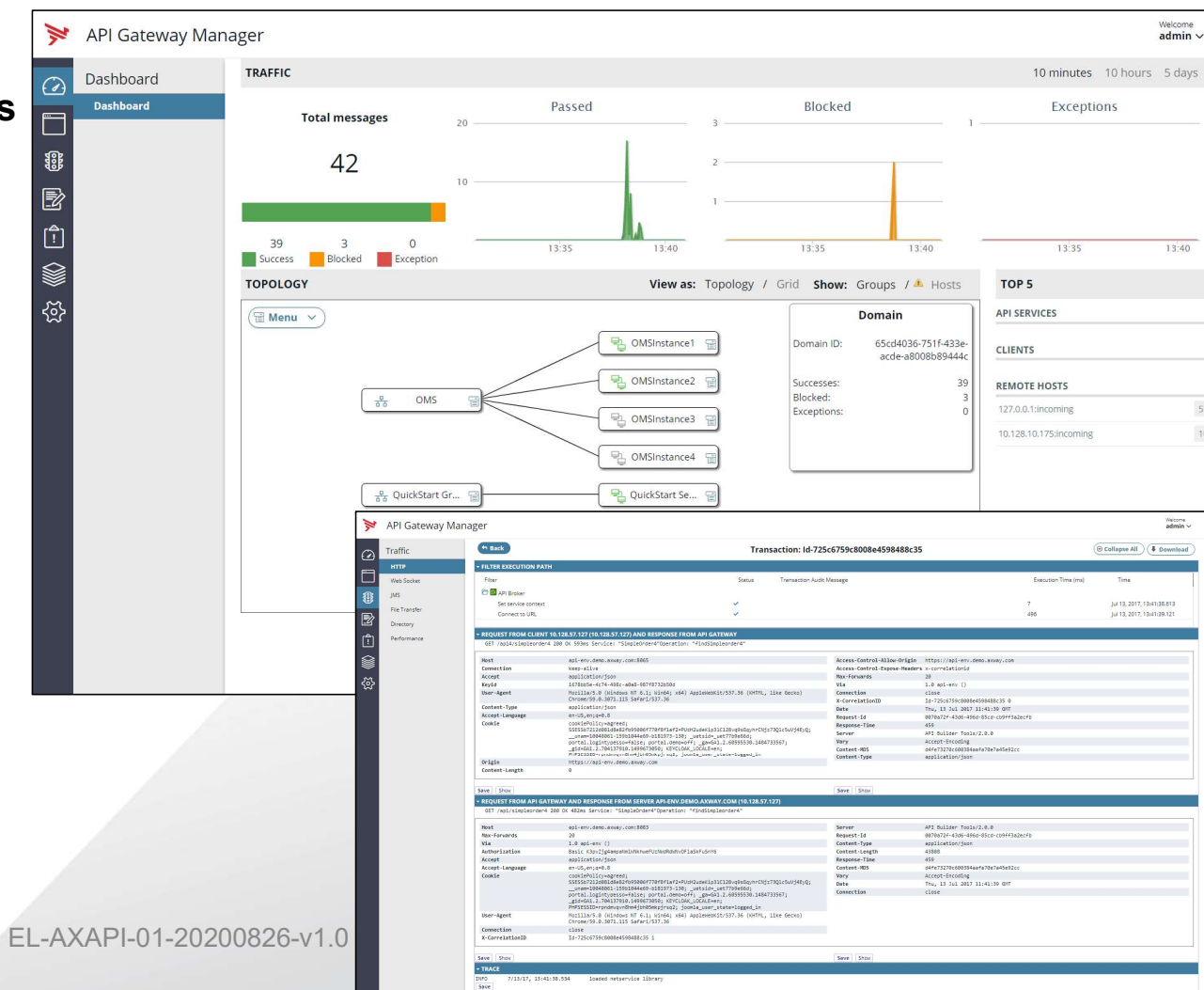
Introdução



Cockpit do Administrador “API Gateway Manager”



- Interface única para Admin & Oper
- Verifique toda saúde da infraestrutura, dos servidores na UI
 - Atividades
 - Cluster & Processos
 - KPI
- Monitorar todo tráfego, de todos servidores
 - Com *drill-down* e todos os detalhes
 - *Tuning* para confidencialidade e performance
 - Auditoria e rastreabilidade
- Operação
 - Start/stop
 - Deploy
- Configuração de administração (KPS, ajuste de nível de log dinâmico, ...)



API Gateway Manager: ponto de vista técnico

- API Gateway Manager é uma interface de usuário no Admin Node Manager (ANM)
 - API Gateway Manager utiliza totalmente no ANM's APIs
 - API Gateway Manager está no mesmo listener (ex porta) que as API's do ANM
- ANM utiliza Role Based Access Control (RBAC)
 - A autenticação do API Gateway Manager pode ser sobrescrito (ex: integração com AD)
 - API Gateway Manager atende objetivamente
 - Primariamente para administradores e operadores
 - Secundariamente para desenvolvedores e security officers
- A respeito de arquitetura
 - ANM pode ser configurado em alta disponibilidade
 - ANM pode ser implementado no servidor remoto (ex não em DMZ)
 - Escalabilidade não esta no tópico API Gateway Manager

Dashboard

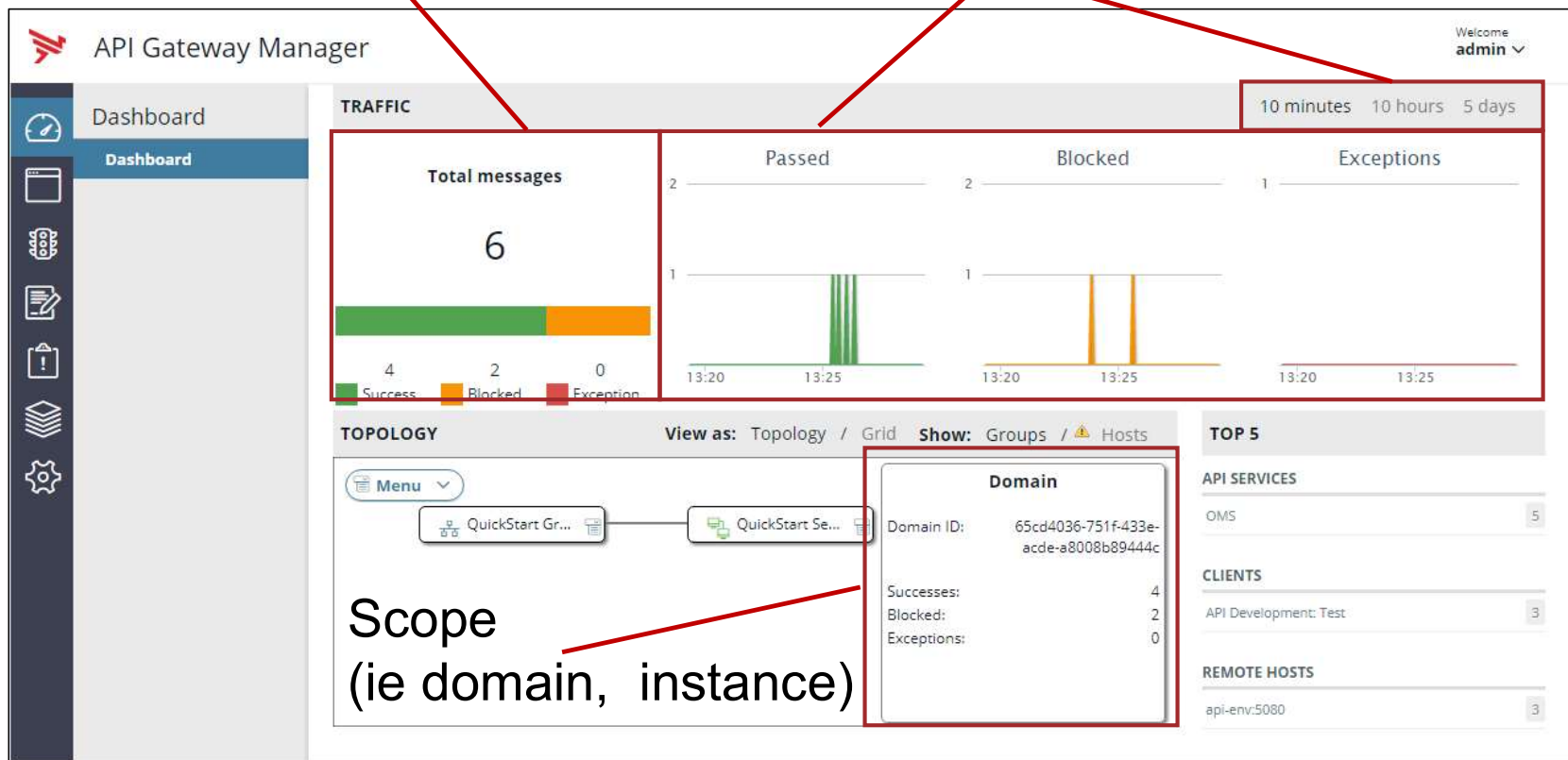
Dashboard: tudo em uma interface



Dashboard: Tráfego

Número total de mensagens
No escopo selecionado

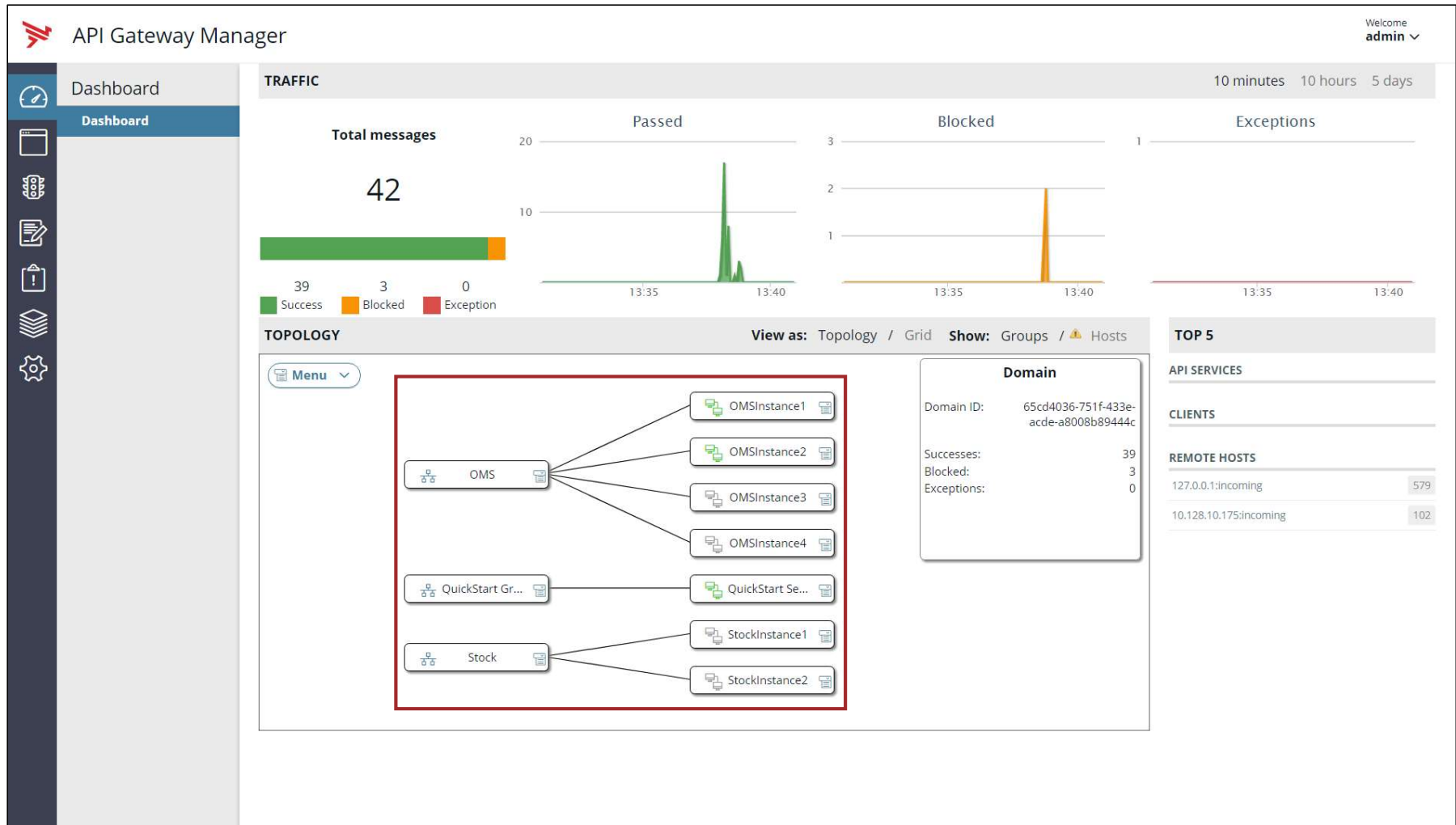
Gráfico de mensagens, no intervalo selecionado



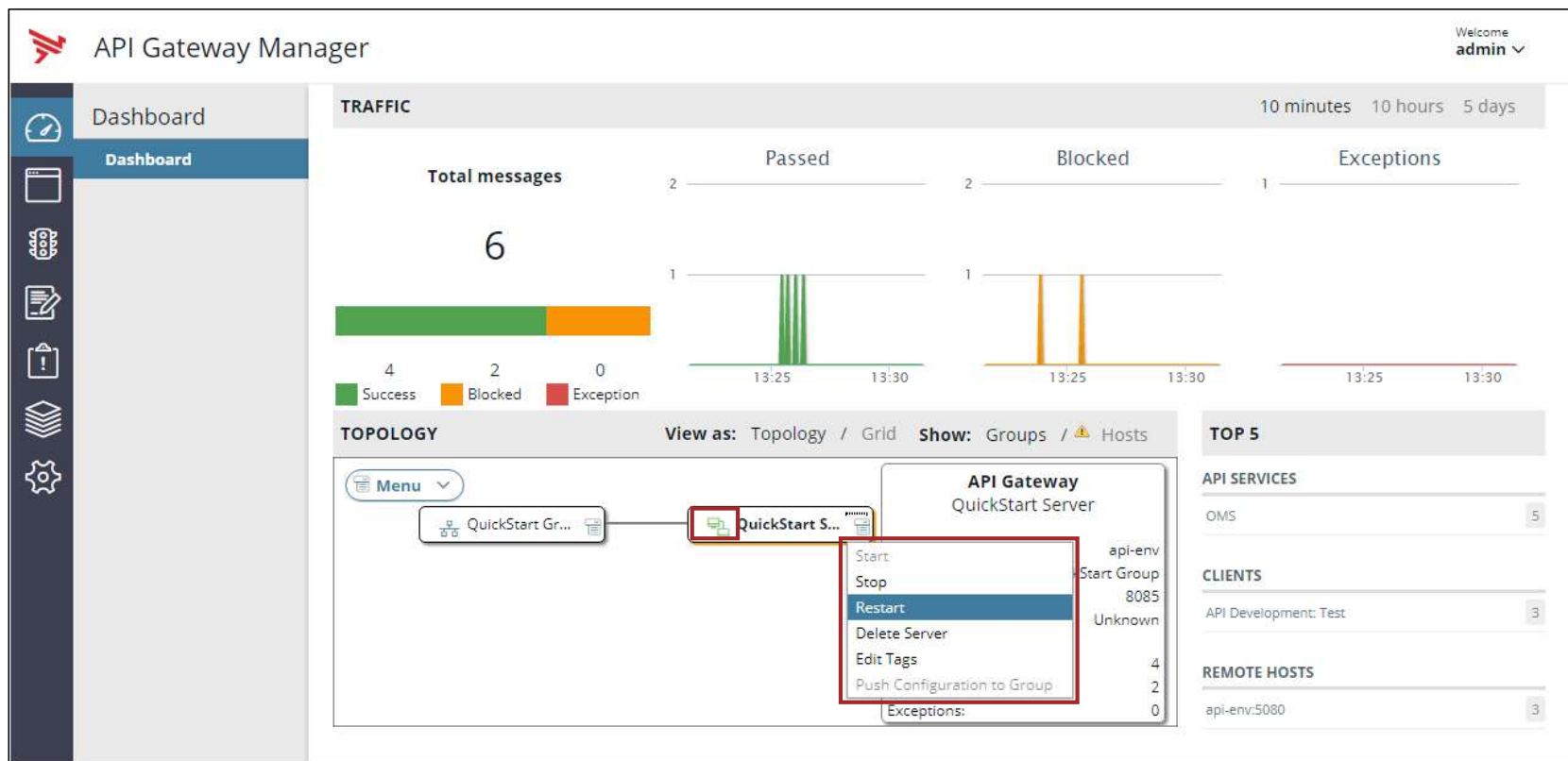
Scope
(ie domain, instance)

Nota: Passed, Blocked, Exceptions são parte do conceito de Policy.

Dashboard: Exemplo de topologia multi-groups, multi-instances

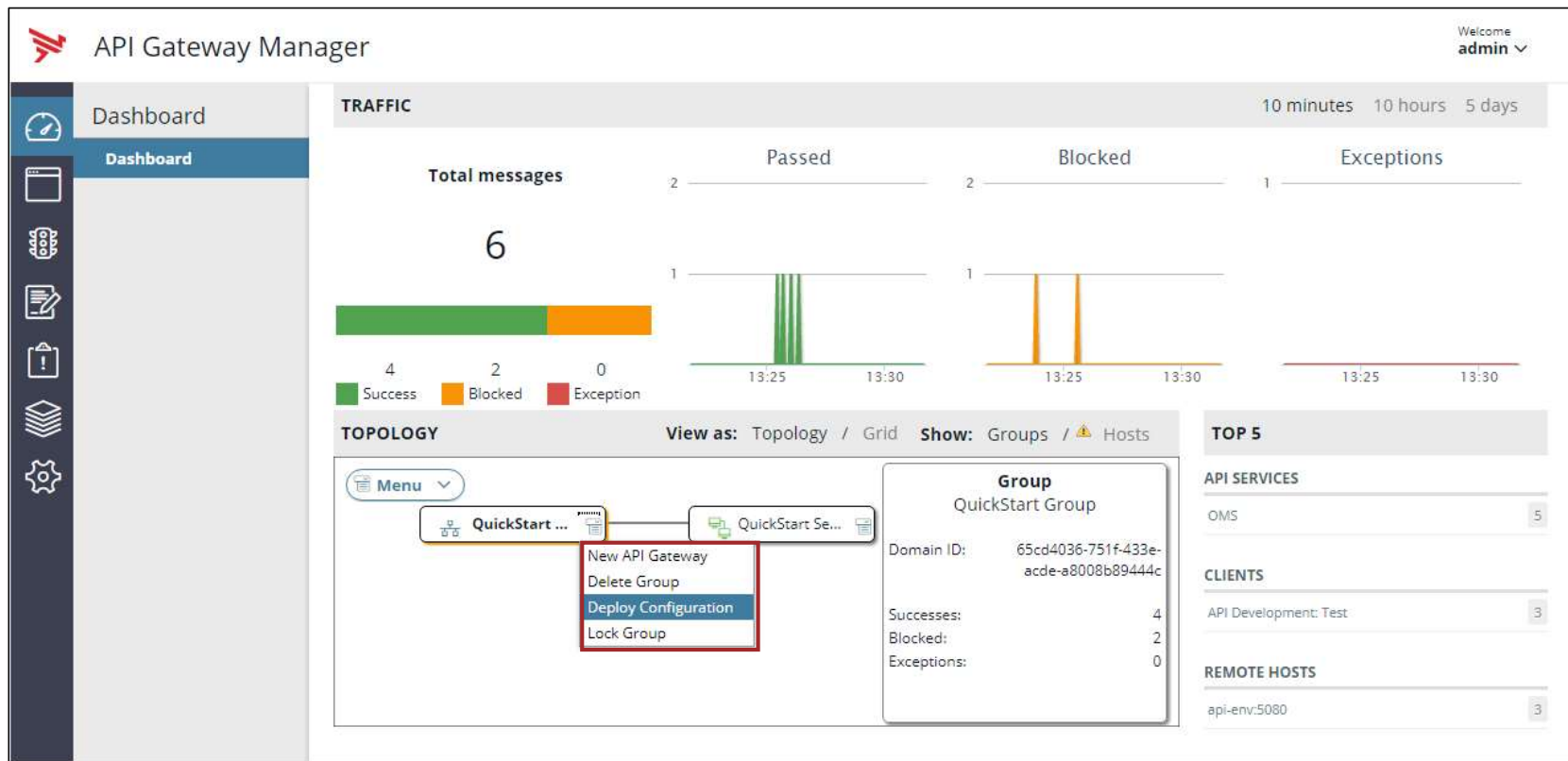


Dashboard: Topology, instance menu



Gerenciamento da instância (start, stop, status)

Dashboard: Topology, group menu



Dashboard: Top 5



TOP 5 API, API Manager client application, server called

Tráfego

Traffic: monitora qualquer atividade, até o menor detalhe

API Gateway Manager Welcome admin

Traffic

HTTP

Web Socket

JMS

File Transfer

Directory

Performance

Filter + **Apply**

GROUPS AND SERVERS **All Servers**

REQUEST FROM **Client**

MAX RESULTS PER SERVER **1000**

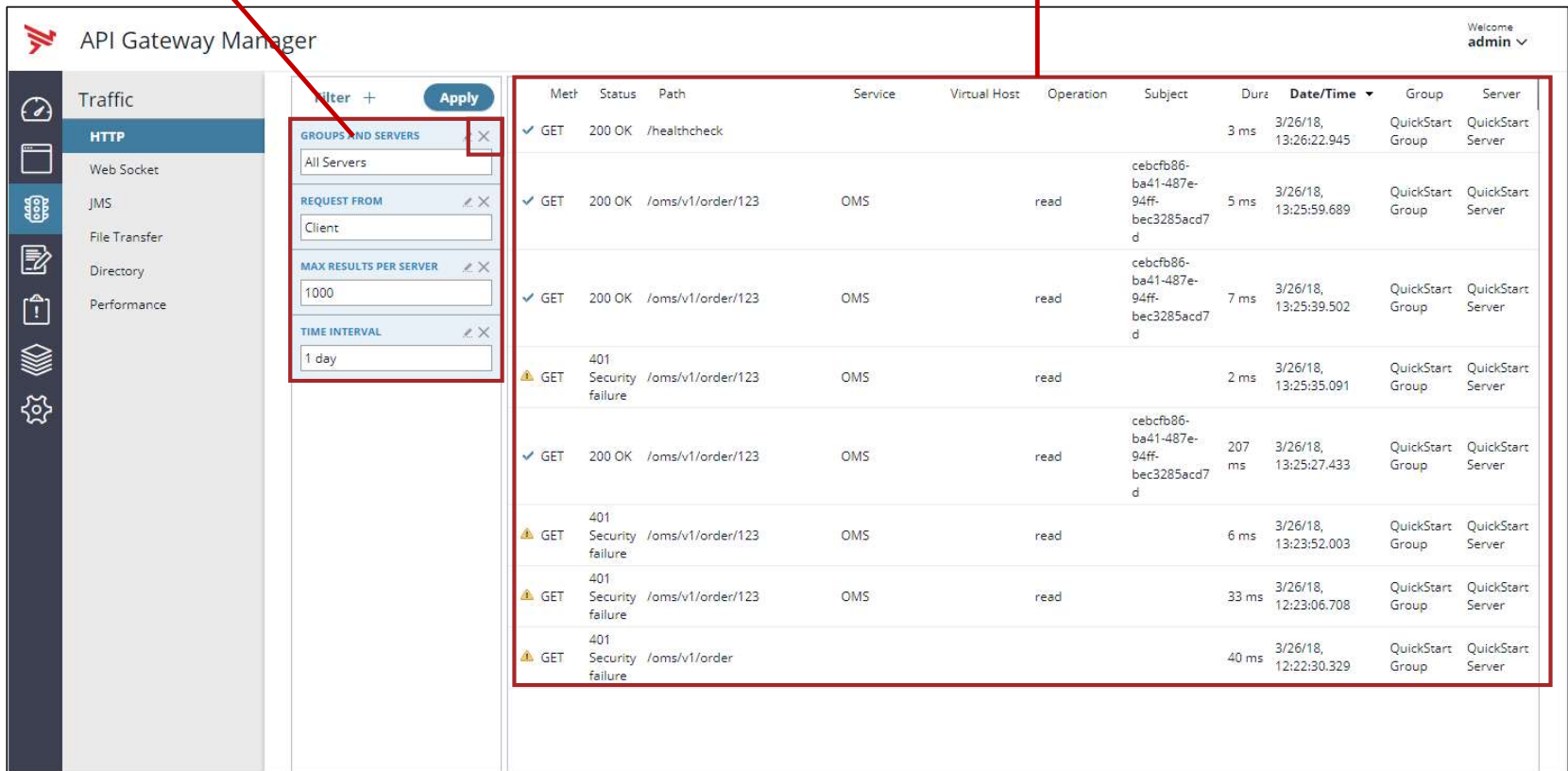
TIME INTERVAL **1 day**

Met	Status	Path	Service	Virtual Host	Operation	Subject	Dura	Date/Time	Group	Server
✓ GET	200 OK	/healthcheck					3 ms	3/26/18, 13:26:22.945	QuickStart Group	QuickStart Server
✓ GET	200 OK	/oms/v1/order/123	Oms		read	cebcfb86-ba41-487e-94ff-bec3285acd7d	5 ms	3/26/18, 13:25:59.689	QuickStart Group	QuickStart Server
✓ GET	200 OK	/oms/v1/order/123	Oms		read	cebcfb86-ba41-487e-94ff-bec3285acd7d	7 ms	3/26/18, 13:25:39.502	QuickStart Group	QuickStart Server
⚠ GET	401 Security failure	/oms/v1/order/123	Oms		read		2 ms	3/26/18, 13:25:35.091	QuickStart Group	QuickStart Server
✓ GET	200 OK	/oms/v1/order/123	Oms		read	cebcfb86-ba41-487e-94ff-bec3285acd7d	207 ms	3/26/18, 13:25:27.433	QuickStart Group	QuickStart Server
⚠ GET	401 Security failure	/oms/v1/order/123	Oms		read		6 ms	3/26/18, 13:23:52.003	QuickStart Group	QuickStart Server
⚠ GET	401 Security failure	/oms/v1/order/123	Oms		read		33 ms	3/26/18, 12:23:06.708	QuickStart Group	QuickStart Server
⚠ GET	401 Security failure	/oms/v1/order/123	Oms		read		40 ms	3/26/18, 12:22:30.329	QuickStart Group	QuickStart Server

Traffic: http transaction list

Filter panel

Transaction list
Click on one for detail



The screenshot displays the API Gateway Manager interface. On the left, a sidebar contains navigation icons for Traffic, Web Socket, JMS, File Transfer, Directory, and Performance. The 'Traffic' section is active, showing a list of transaction types: HTTP, Web Socket, JMS, File Transfer, Directory, and Performance. The 'HTTP' section is selected, and a filter panel is visible on the right. The filter panel includes a 'Filter +' button, an 'Apply' button, and several filter criteria: 'GROUPS AND SERVERS' (set to 'All Servers'), 'REQUEST FROM' (set to 'Client'), 'MAX RESULTS PER SERVER' (set to '1000'), and 'TIME INTERVAL' (set to '1 day').

The main area displays a table of HTTP transactions. The table has columns: Method, Status, Path, Service, Virtual Host, Operation, Subject, Durs, Date/Time, Group, and Server. The transactions are listed as follows:

Method	Status	Path	Service	Virtual Host	Operation	Subject	Durs	Date/Time	Group	Server
✓ GET	200 OK	/healthcheck					3 ms	3/26/18, 13:26:22.945	QuickStart Group	QuickStart Server
✓ GET	200 OK	/oms/v1/order/123	Oms		read	cebcfb86-ba41-487e-94ff-bec3285acd7d	5 ms	3/26/18, 13:25:59.689	QuickStart Group	QuickStart Server
✓ GET	200 OK	/oms/v1/order/123	Oms		read	cebcfb86-ba41-487e-94ff-bec3285acd7d	7 ms	3/26/18, 13:25:39.502	QuickStart Group	QuickStart Server
⚠ GET	401 Security failure	/oms/v1/order/123	Oms		read		2 ms	3/26/18, 13:25:35.091	QuickStart Group	QuickStart Server
✓ GET	200 OK	/oms/v1/order/123	Oms		read	cebcfb86-ba41-487e-94ff-bec3285acd7d	207 ms	3/26/18, 13:25:27.433	QuickStart Group	QuickStart Server
⚠ GET	401 Security failure	/oms/v1/order/123	Oms		read		6 ms	3/26/18, 13:23:52.003	QuickStart Group	QuickStart Server
⚠ GET	401 Security failure	/oms/v1/order/123	Oms		read		33 ms	3/26/18, 12:23:06.708	QuickStart Group	QuickStart Server
⚠ GET	401 Security failure	/oms/v1/order/123	Oms		read		40 ms	3/26/18, 12:22:30.329	QuickStart Group	QuickStart Server

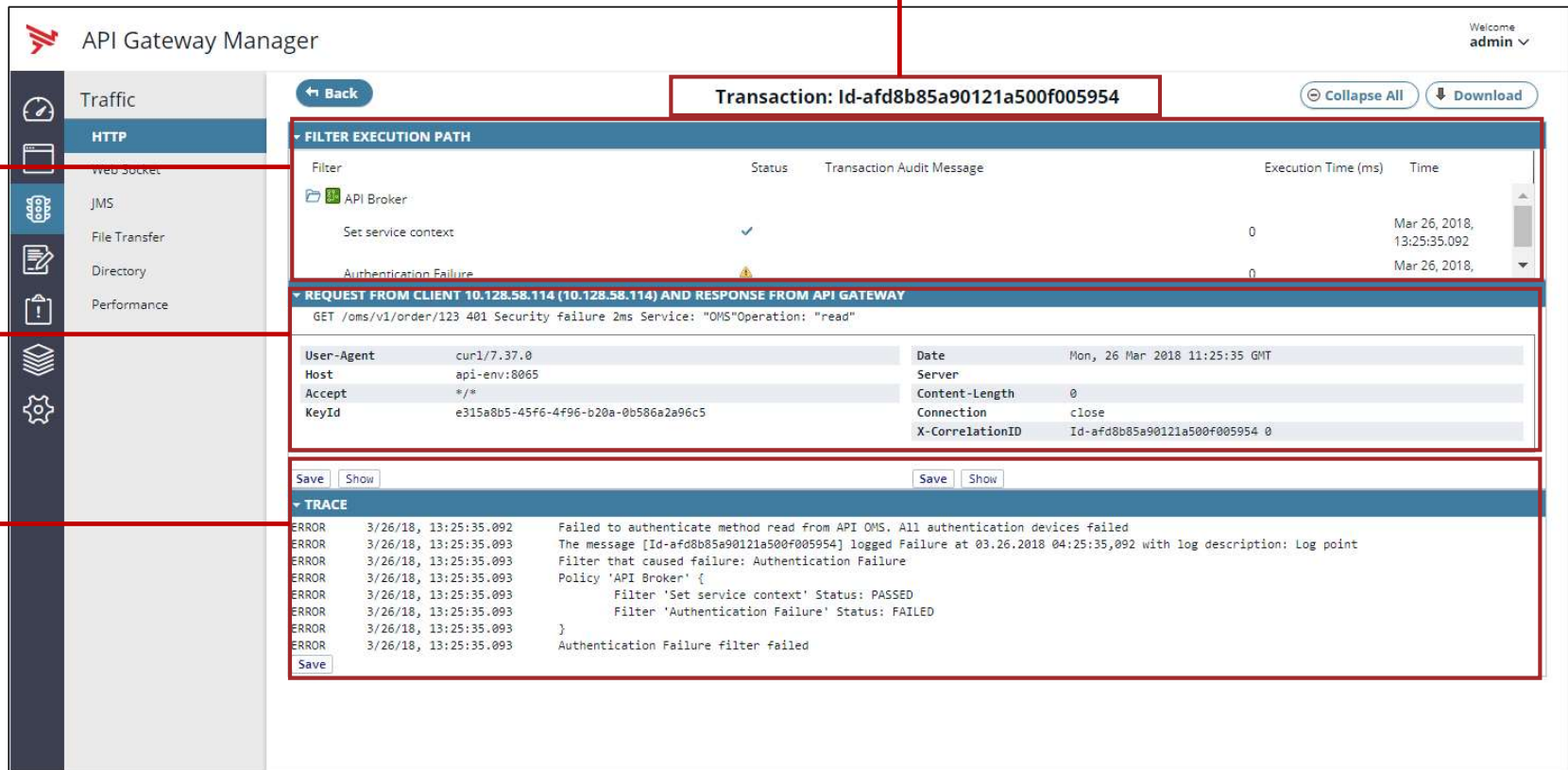
Traffic: http transaction detail

Unique Transaction id, set by API Gateway

Execution steps

HTTP requests dump

Traces



API Gateway Manager

Welcome admin

Transaction: Id-afd8b85a90121a500f005954 [Collapse All] [Download]

FILTER EXECUTION PATH

Filter	Status	Transaction Audit Message	Execution Time (ms)	Time
API Broker				
Set service context	✓		0	Mar 26, 2018, 13:25:35.092
Authentication Failure	⚠		0	Mar 26, 2018, 13:25:35.092

REQUEST FROM CLIENT 10.128.58.114 (10.128.58.114) AND RESPONSE FROM API GATEWAY

GET /oms/v1/order/123 401 Security failure 2ms Service: "OMS"Operation: "read"

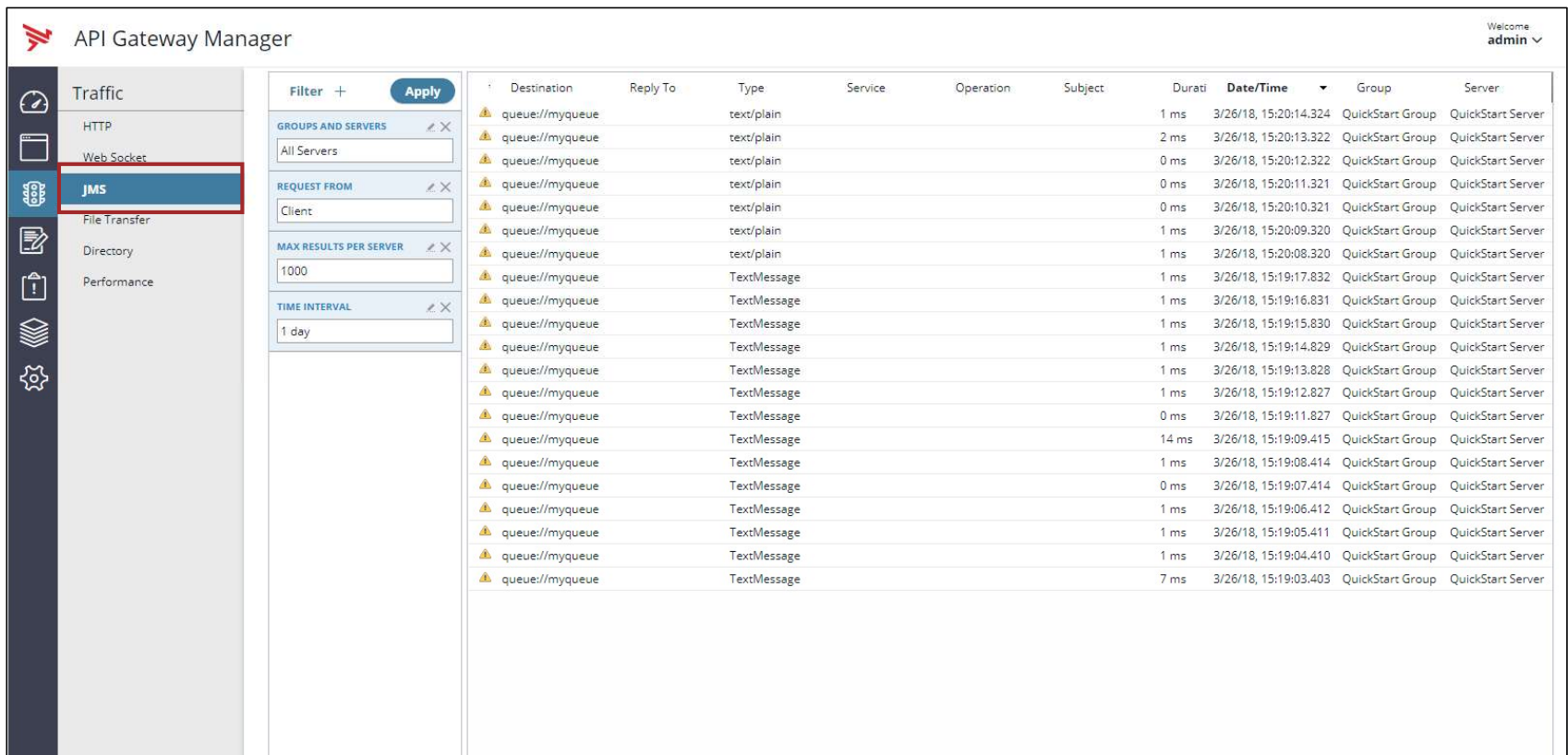
User-Agent	curl/7.37.0	Date	Mon, 26 Mar 2018 11:25:35 GMT
Host	api-env:8065	Server	
Accept	*/*	Content-Length	0
KeyId	e315a8b5-45f6-4f96-b20a-0b586a2a96c5	Connection	close
		X-CorrelationID	Id-afd8b85a90121a500f005954 0

TRACE

Time	Message
3/26/18, 13:25:35.092	Failed to authenticate method read from API OMS. All authentication devices failed
3/26/18, 13:25:35.093	The message [Id-afd8b85a90121a500f005954] logged Failure at 03.26.2018 04:25:35,092 with log description: Log point
3/26/18, 13:25:35.093	Filter that caused failure: Authentication Failure
3/26/18, 13:25:35.093	Policy 'API Broker' {
3/26/18, 13:25:35.093	Filter 'Set service context' Status: PASSED
3/26/18, 13:25:35.093	Filter 'Authentication Failure' Status: FAILED
3/26/18, 13:25:35.093	}
3/26/18, 13:25:35.093	Authentication Failure filter failed

See “Monitor and Troubleshoot” module for complete explanation

Traffic: not only HTTP! Different per protocol



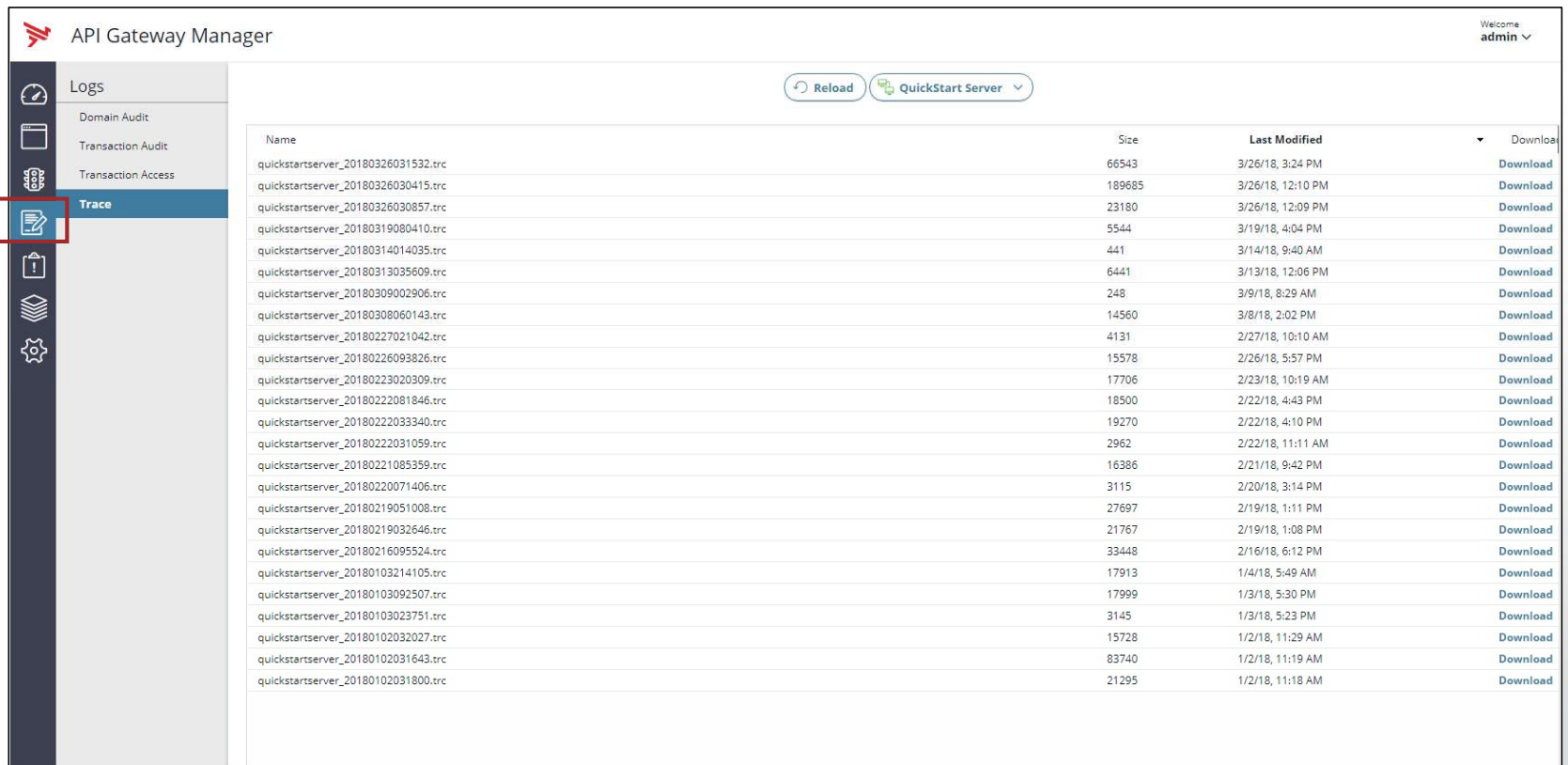
The screenshot shows the API Gateway Manager interface. On the left, a sidebar contains navigation icons and labels: Traffic, HTTP, Web Socket, JMS (highlighted with a red box), File Transfer, Directory, and Performance. The main area displays a table of traffic logs for the JMS protocol. The table has columns for Destination, Reply To, Type, Service, Operation, Subject, Duration, Date/Time, Group, and Server. The logs show a series of messages sent to 'queue://myqueue' with various timestamps and durations. A filter panel on the left of the table allows for filtering by groups and servers, request from, max results per server, and time interval.

Destination	Reply To	Type	Service	Operation	Subject	Duration	Date/Time	Group	Server
queue://myqueue		text/plain				1 ms	3/26/18, 15:20:14.324	QuickStart Group	QuickStart Server
queue://myqueue		text/plain				2 ms	3/26/18, 15:20:13.322	QuickStart Group	QuickStart Server
queue://myqueue		text/plain				0 ms	3/26/18, 15:20:12.322	QuickStart Group	QuickStart Server
queue://myqueue		text/plain				0 ms	3/26/18, 15:20:11.321	QuickStart Group	QuickStart Server
queue://myqueue		text/plain				0 ms	3/26/18, 15:20:10.321	QuickStart Group	QuickStart Server
queue://myqueue		text/plain				1 ms	3/26/18, 15:20:09.320	QuickStart Group	QuickStart Server
queue://myqueue		text/plain				1 ms	3/26/18, 15:20:08.320	QuickStart Group	QuickStart Server
queue://myqueue		TextMessage				1 ms	3/26/18, 15:19:17.832	QuickStart Group	QuickStart Server
queue://myqueue		TextMessage				1 ms	3/26/18, 15:19:16.831	QuickStart Group	QuickStart Server
queue://myqueue		TextMessage				1 ms	3/26/18, 15:19:15.830	QuickStart Group	QuickStart Server
queue://myqueue		TextMessage				1 ms	3/26/18, 15:19:14.829	QuickStart Group	QuickStart Server
queue://myqueue		TextMessage				1 ms	3/26/18, 15:19:13.828	QuickStart Group	QuickStart Server
queue://myqueue		TextMessage				1 ms	3/26/18, 15:19:12.827	QuickStart Group	QuickStart Server
queue://myqueue		TextMessage				0 ms	3/26/18, 15:19:11.827	QuickStart Group	QuickStart Server
queue://myqueue		TextMessage				14 ms	3/26/18, 15:19:09.415	QuickStart Group	QuickStart Server
queue://myqueue		TextMessage				1 ms	3/26/18, 15:19:08.414	QuickStart Group	QuickStart Server
queue://myqueue		TextMessage				0 ms	3/26/18, 15:19:07.414	QuickStart Group	QuickStart Server
queue://myqueue		TextMessage				1 ms	3/26/18, 15:19:06.412	QuickStart Group	QuickStart Server
queue://myqueue		TextMessage				1 ms	3/26/18, 15:19:05.411	QuickStart Group	QuickStart Server
queue://myqueue		TextMessage				1 ms	3/26/18, 15:19:04.410	QuickStart Group	QuickStart Server
queue://myqueue		TextMessage				7 ms	3/26/18, 15:19:03.403	QuickStart Group	QuickStart Server

Do not miss errors!

Logs and Events

Logs: access all logs, from every servers



API Gateway Manager

Welcome admin

Logs

- Domain Audit
- Transaction Audit
- Transaction Access
- Trace**

Reload QuickStart Server

Name	Size	Last Modified	Download
quickstartserver_20180326031532.trc	66543	3/26/18, 3:24 PM	Download
quickstartserver_20180326030415.trc	189685	3/26/18, 12:10 PM	Download
quickstartserver_20180326030857.trc	23180	3/26/18, 12:09 PM	Download
quickstartserver_20180319080410.trc	5544	3/19/18, 4:04 PM	Download
quickstartserver_20180314014035.trc	441	3/14/18, 9:40 AM	Download
quickstartserver_20180313035609.trc	6441	3/13/18, 12:06 PM	Download
quickstartserver_20180309002906.trc	248	3/9/18, 8:29 AM	Download
quickstartserver_20180308060143.trc	14560	3/8/18, 2:02 PM	Download
quickstartserver_20180227021042.trc	4131	2/27/18, 10:10 AM	Download
quickstartserver_20180226093826.trc	15578	2/26/18, 5:57 PM	Download
quickstartserver_20180223020309.trc	17706	2/23/18, 10:19 AM	Download
quickstartserver_20180222081846.trc	18500	2/22/18, 4:43 PM	Download
quickstartserver_20180222033340.trc	19270	2/22/18, 4:10 PM	Download
quickstartserver_20180222031059.trc	2962	2/22/18, 11:11 AM	Download
quickstartserver_20180221085359.trc	16386	2/21/18, 9:42 PM	Download
quickstartserver_20180220071406.trc	3115	2/20/18, 3:14 PM	Download
quickstartserver_20180219051008.trc	27697	2/19/18, 1:11 PM	Download
quickstartserver_20180219032646.trc	21767	2/19/18, 1:08 PM	Download
quickstartserver_20180216095524.trc	33448	2/16/18, 6:12 PM	Download
quickstartserver_20180103214105.trc	17913	1/4/18, 5:49 AM	Download
quickstartserver_20180103092507.trc	17999	1/3/18, 5:30 PM	Download
quickstartserver_20180103023751.trc	3145	1/3/18, 5:23 PM	Download
quickstartserver_20180102032027.trc	15728	1/2/18, 11:29 AM	Download
quickstartserver_20180102031643.trc	83740	1/2/18, 11:19 AM	Download
quickstartserver_20180102031800.trc	21295	1/2/18, 11:18 AM	Download

Events: review all alerts

API Gateway Manager Welcome admin

Events QuickStart Server

Transaction audit

Alerts

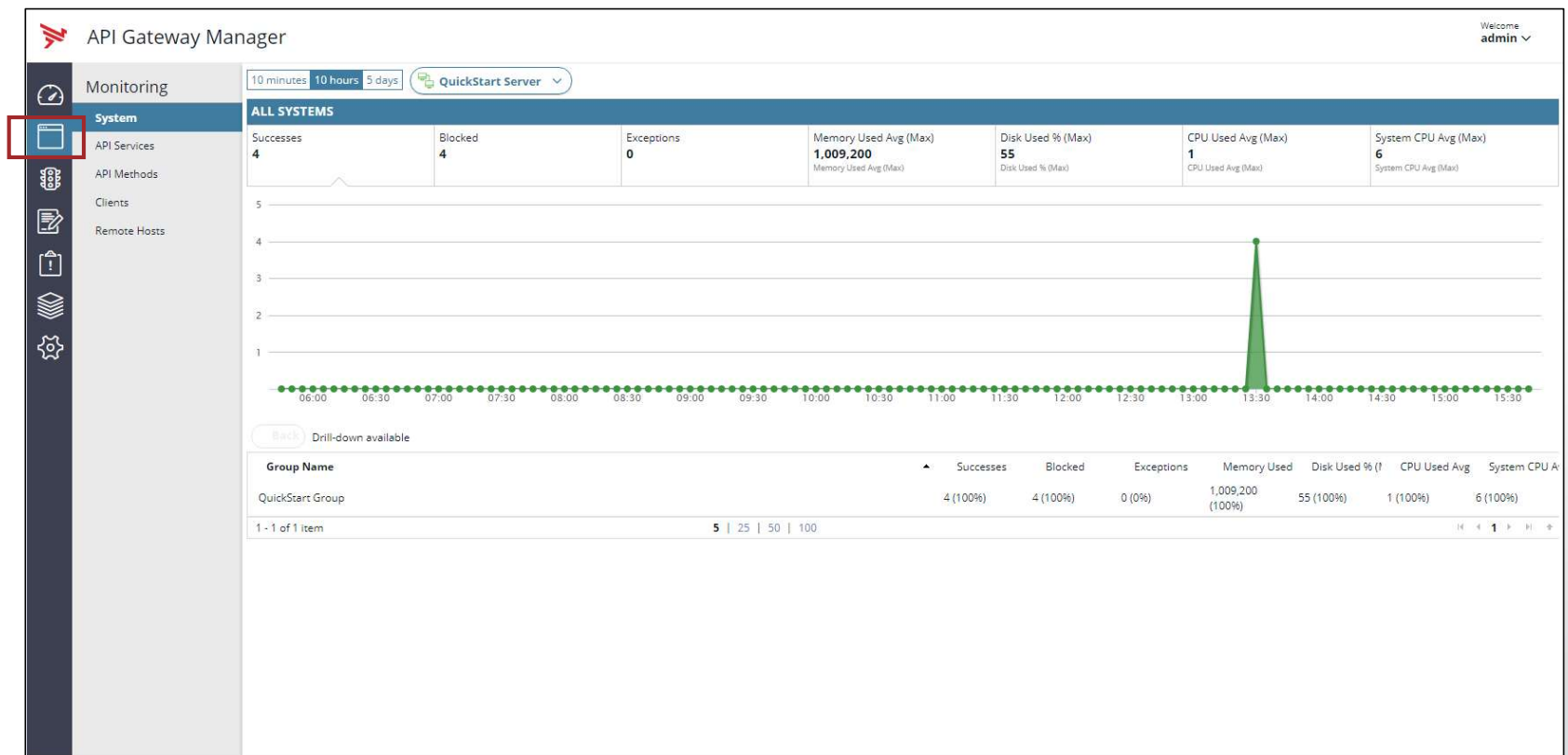
SLA Alerts

Message Id	Message	Source	API Gateway	Group	Date/Time
Id-afd8b85a90121a500f005954	Log point	Authentication Failure	QuickStart Server	QuickStart Group	3/26/18, 13:25:35.092
Id-48d8b85a1112ecdac5b463e4	Log point	Authentication Failure	QuickStart Server	QuickStart Group	3/26/18, 13:23:52.004
Id-0acab85a2b023b2957685f9f	Log point	Authentication Failure	QuickStart Server	QuickStart Group	3/26/18, 12:23:06.739
Id-e6c9b85a01024e44d10958e7	Log point	Authentication Failure	QuickStart Server	QuickStart Group	3/26/18, 12:22:30.360

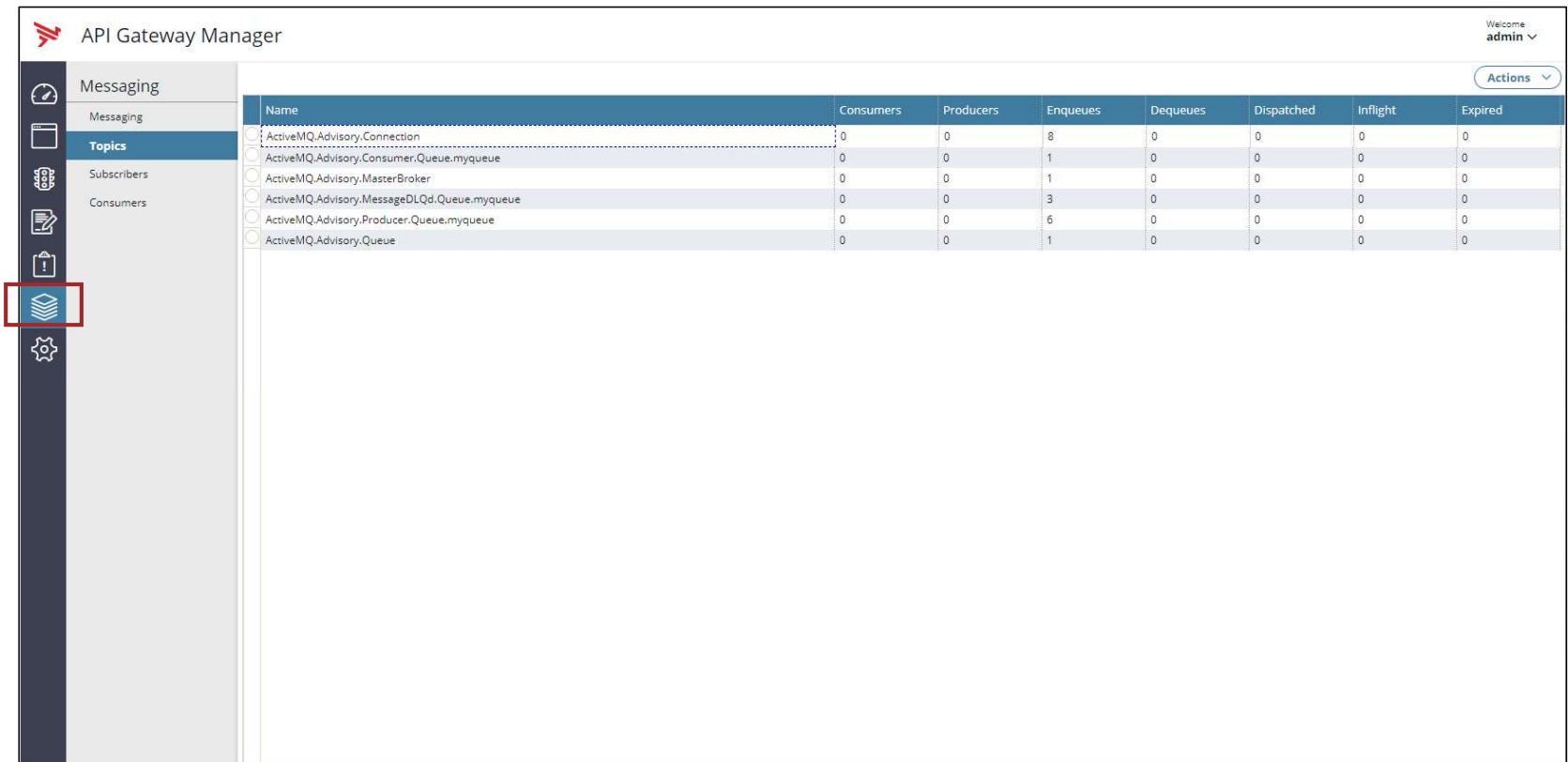
1 - 4 of 4 items 10 | 25 | 50 | 100

Outras funcionalidades

Monitoring: dashboards de atividades em tempo real



Messaging: gerenciamento embedded Active-MQ

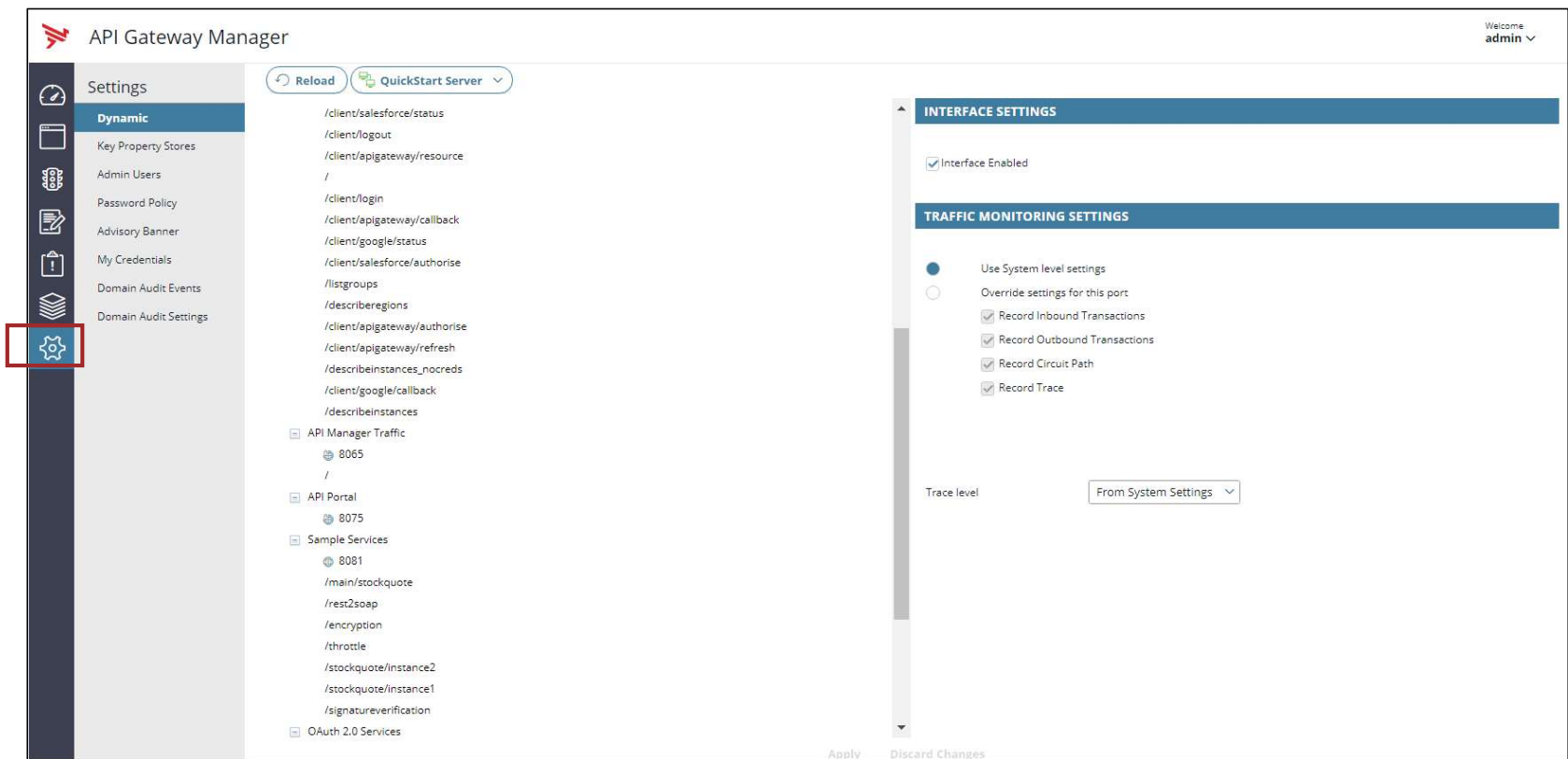


The screenshot shows the API Gateway Manager interface. On the left, a sidebar contains icons for Messaging, Topics, Subscribers, Consumers, and a red box highlighting the Embedded ActiveMQ icon. The main area displays a table of ActiveMQ topics with columns for Name, Consumers, Producers, Enqueues, Dequeues, Dispatched, Inflight, and Expired.

Name	Consumers	Producers	Enqueues	Dequeues	Dispatched	Inflight	Expired
ActiveMQ.Advisory.Connection	0	0	8	0	0	0	0
ActiveMQ.Advisory.Consumer.Queue.myqueue	0	0	1	0	0	0	0
ActiveMQ.Advisory.MasterBroker	0	0	1	0	0	0	0
ActiveMQ.Advisory.MessageDLQd.Queue.myqueue	0	0	3	0	0	0	0
ActiveMQ.Advisory.Producer.Queue.myqueue	0	0	6	0	0	0	0
ActiveMQ.Advisory.Queue	0	0	1	0	0	0	0

Of course, embedded Active-MQ should be activated
(In Policy Studio, Server Settings)

Settings: configurações dinâmicas



API Gateway Manager

Settings

Dynamic

- Key Property Stores
- Admin Users
- Password Policy
- Advisory Banner
- My Credentials
- Domain Audit Events
- Domain Audit Settings

Reload QuickStart Server

- /client/salesforce/status
- /client/logout
- /client/apigateway/resource
- /
- /client/login
- /client/apigateway/callback
- /client/google/status
- /client/salesforce/authorise
- /listgroups
- /describeinstances
- /client/apigateway/authorise
- /client/apigateway/refresh
- /describeinstances_nocreds
- /client/google/callback
- /describeinstances
- API Manager Traffic
 - 8065
 - /
- API Portal
 - 8075
- Sample Services
 - 8081
 - /main/stockquote
 - /rest2soap
 - /encryption
 - /throttle
 - /stockquote/instance2
 - /stockquote/instance1
 - /signatureverification
- OAuth 2.0 Services

INTERFACE SETTINGS

☒ Interface Enabled

TRAFFIC MONITORING SETTINGS

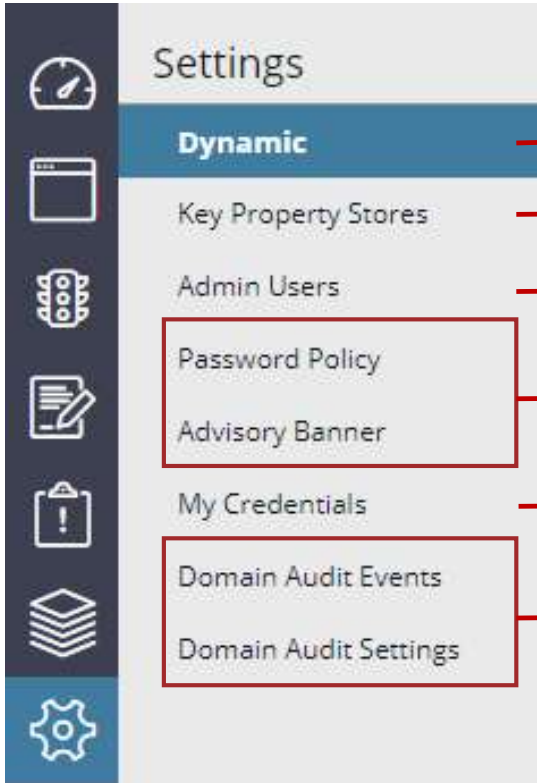
☒ Use System level settings
☐ Override settings for this port

- ☒ Record Inbound Transactions
- ☒ Record Outbound Transactions
- ☒ Record Circuit Path
- ☒ Record Trace

Trace level: From System Settings

Apply Discard Changes

Settings: detalhes



The screenshot shows a settings menu with a dark sidebar on the left containing icons for various functions. The main menu is light gray. The 'Dynamic' option is highlighted in blue. Red lines connect specific menu items to their descriptions on the right. A red box highlights the 'Password Policy' and 'Advisory Banner' items, and another red box highlights the 'Domain Audit Events' and 'Domain Audit Settings' items.

Settings Item	Description
Dynamic	Dynamic log level
Key Property Stores	KPS (ie Cassandra tables)
Admin Users	User management (if local)
Password Policy	Security configuration
Advisory Banner	
My Credentials	Change your password
Domain Audit Events	Domain log configuration
Domain Audit Settings	





Lab – Acessando o Gateway Manager

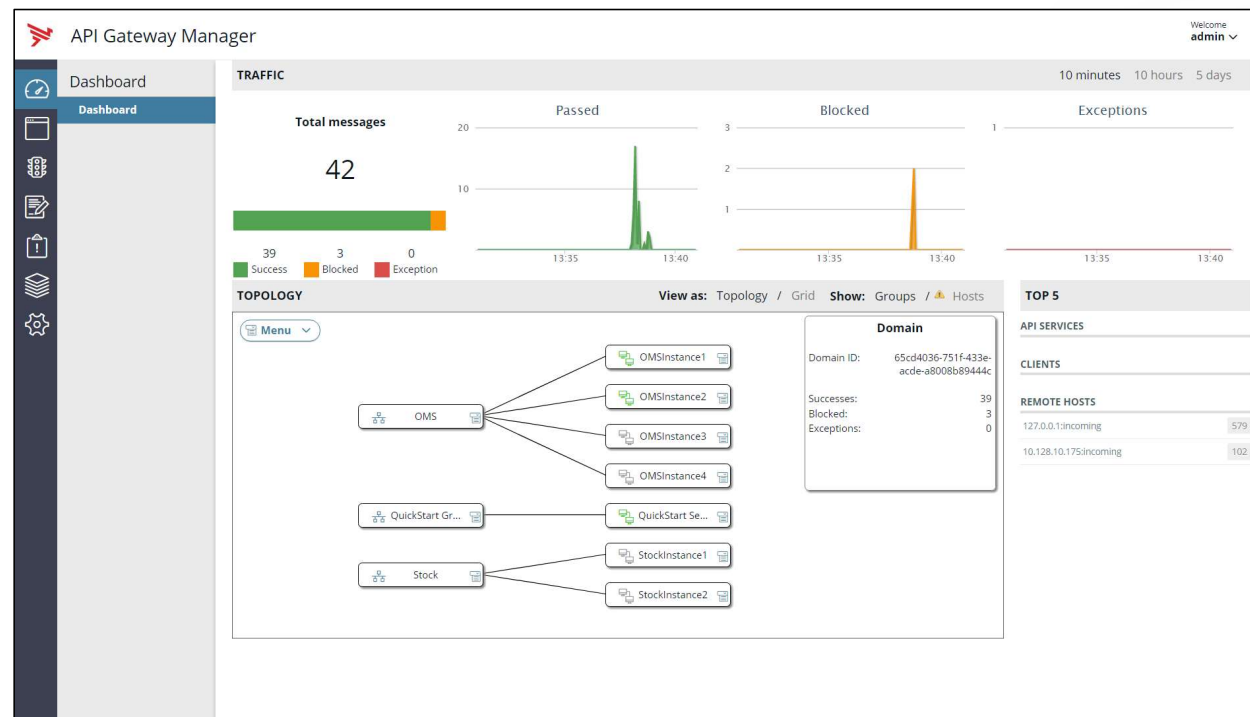


Verificação Ambiente

Verificação padrão

Verificação básica – está no ar?!

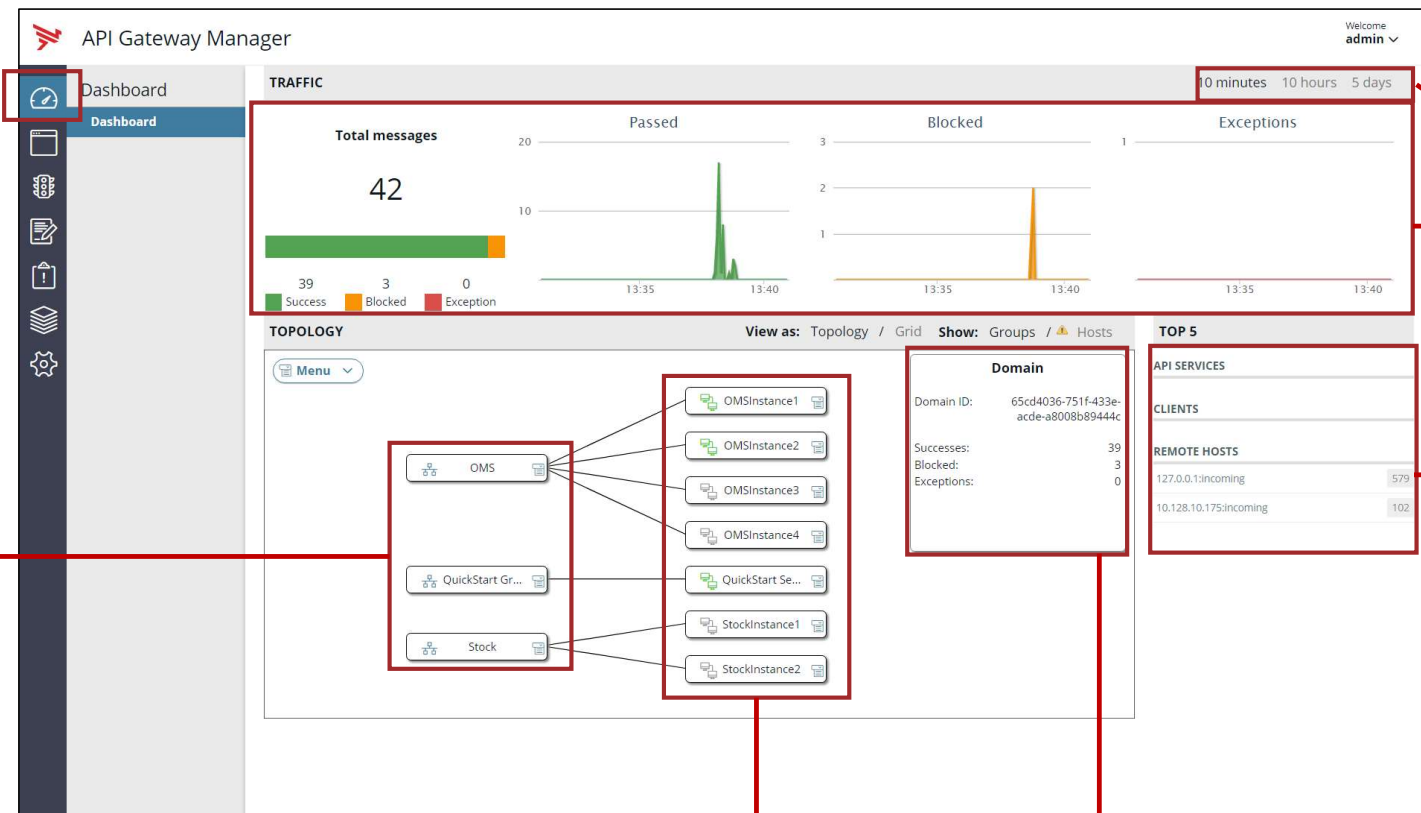
- Como um administrador de TI, o mais importante é ter o seu Sistema trabalhando corretamente
- Vamos entender o que significa trabalhando corretamente e como verificar



O que é uma infraestrutura saudável?

- Todos os processos em execução
 - Node Manager / Admin Node Manager
 - Instances
 - Componentes adicionais (API Portal, Embedded Analytics, ...)
- Consumo dos recursos conforme esperado
- Tráfego processado “normalmente”

Verificação básica – use o API Gateway Manager! Inicialmente via "Dashboard"!



Traffic group and summary (duration)

Top 5 KPI

Group status (configuration consistency)

Instance status (process)

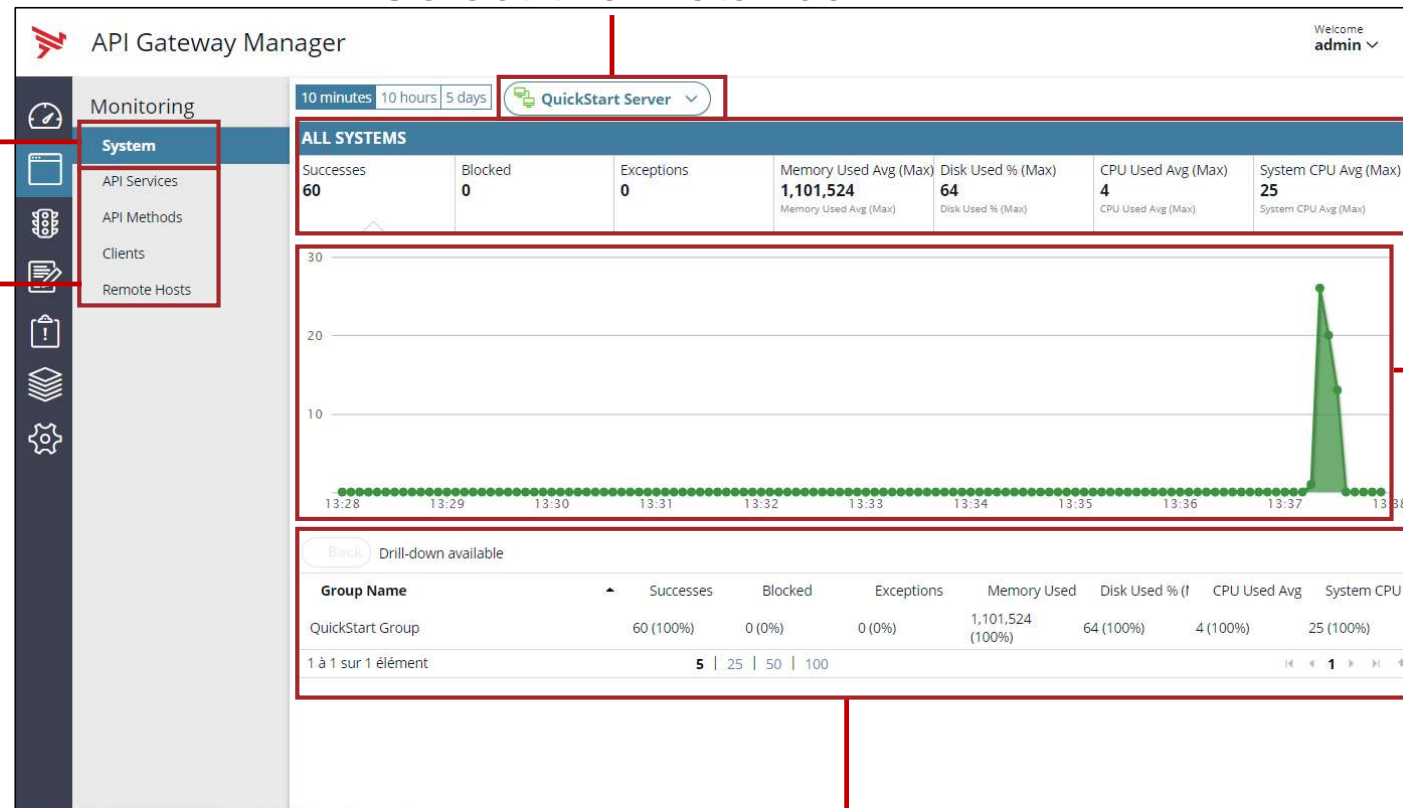
KPI, per Instance or Group selected

Verificação básica – use o API Gateway Manager!

Monitoring tab – utilizado para recursos ("System")

Select the instance

Resources
Traffic aspect



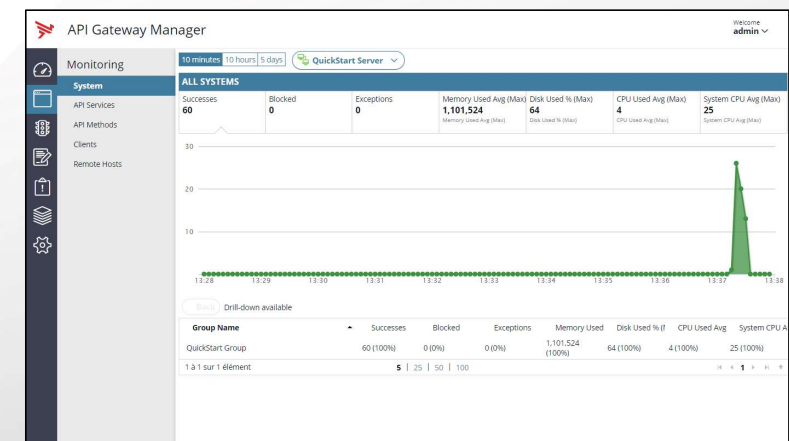
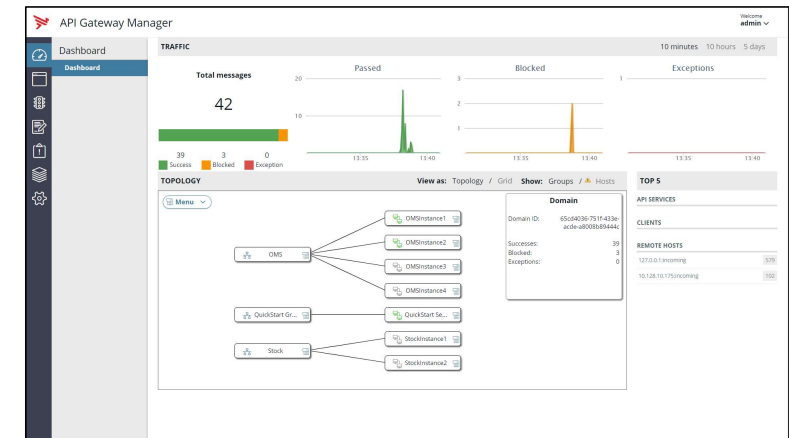
Key
resources

Key
resources

Data group by aspect

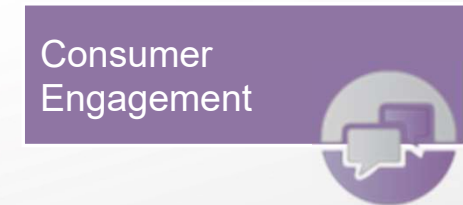
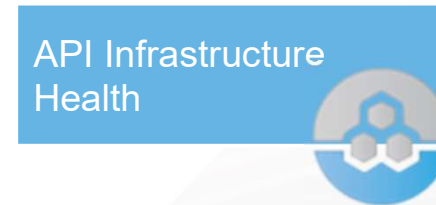
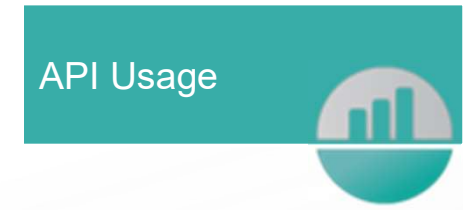
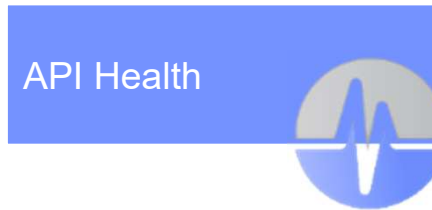
Verificação básica : está rodando

- Utilizando-se a primeira e segunda aba permite ao administrador saber se o processamento de tráfego esta ocorrendo
 - Processo da Instância
 - Visão geral do tráfego
 - Recursos
- Nota: isto não inclui o API Portal ou Embedded Analytics



Standard assessment : it runs "normally"

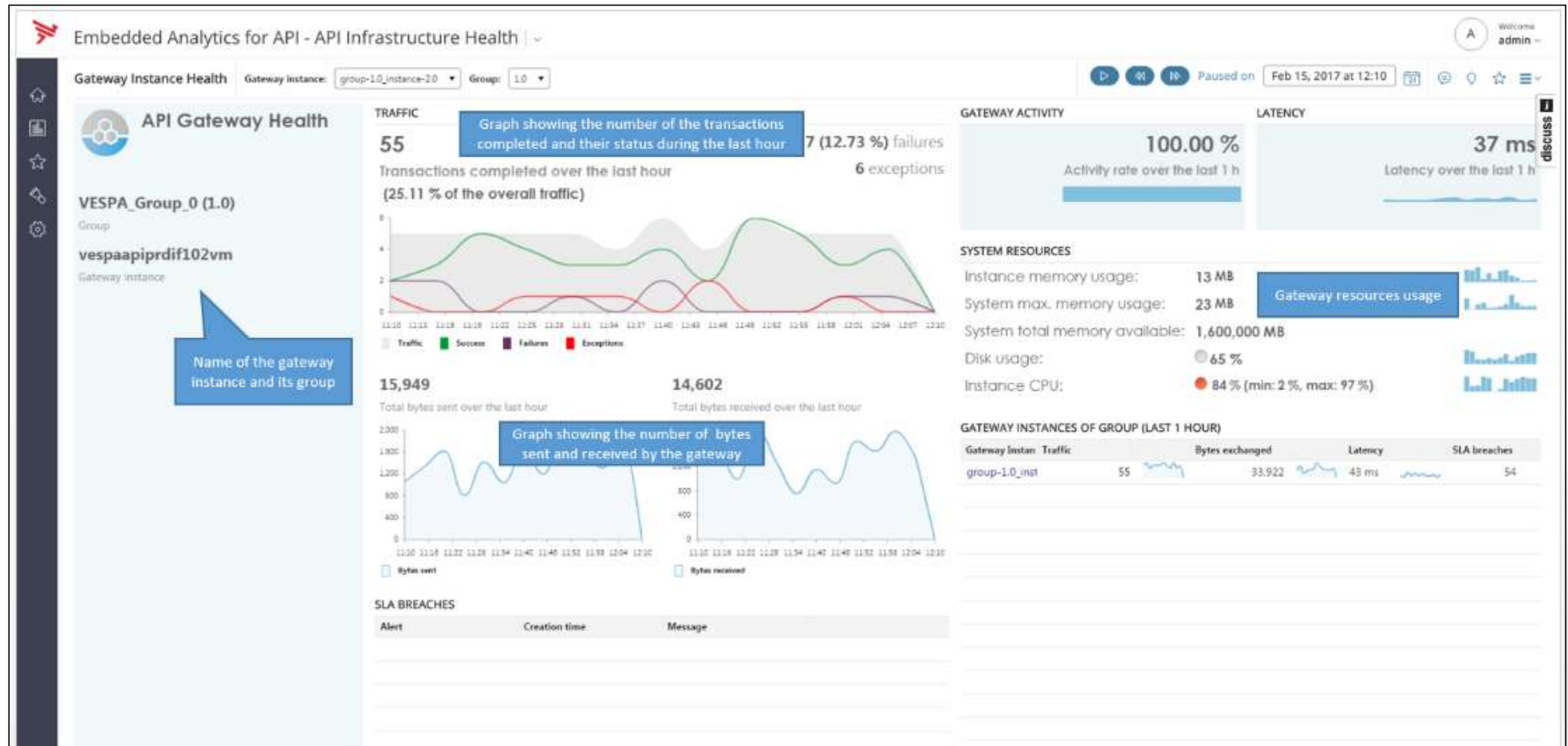
- "Normally" to be defined by Business, Dev and Admin
 - Definition is obvious and very hard to do at the same time
- Take advantage of Embedded Analytics!
- One dashboard per persona
 - API Admin
API Health
 - API Product Manager
API Usage
 - Relationship Manager
Consumer Engagement
 - API Infrastructure Admin
API Infrastructure Health



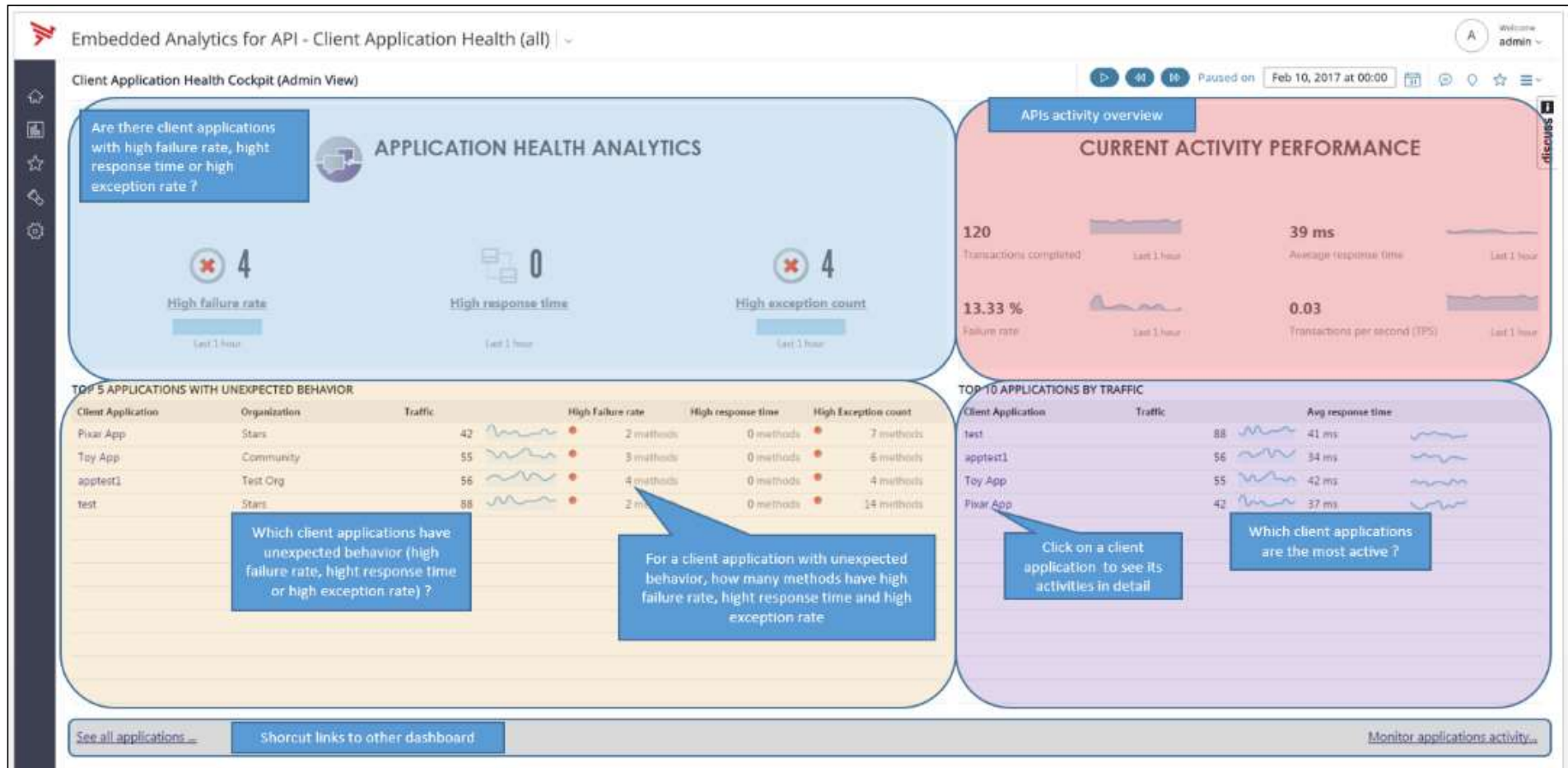
API Infrastructure Health - Cockpit



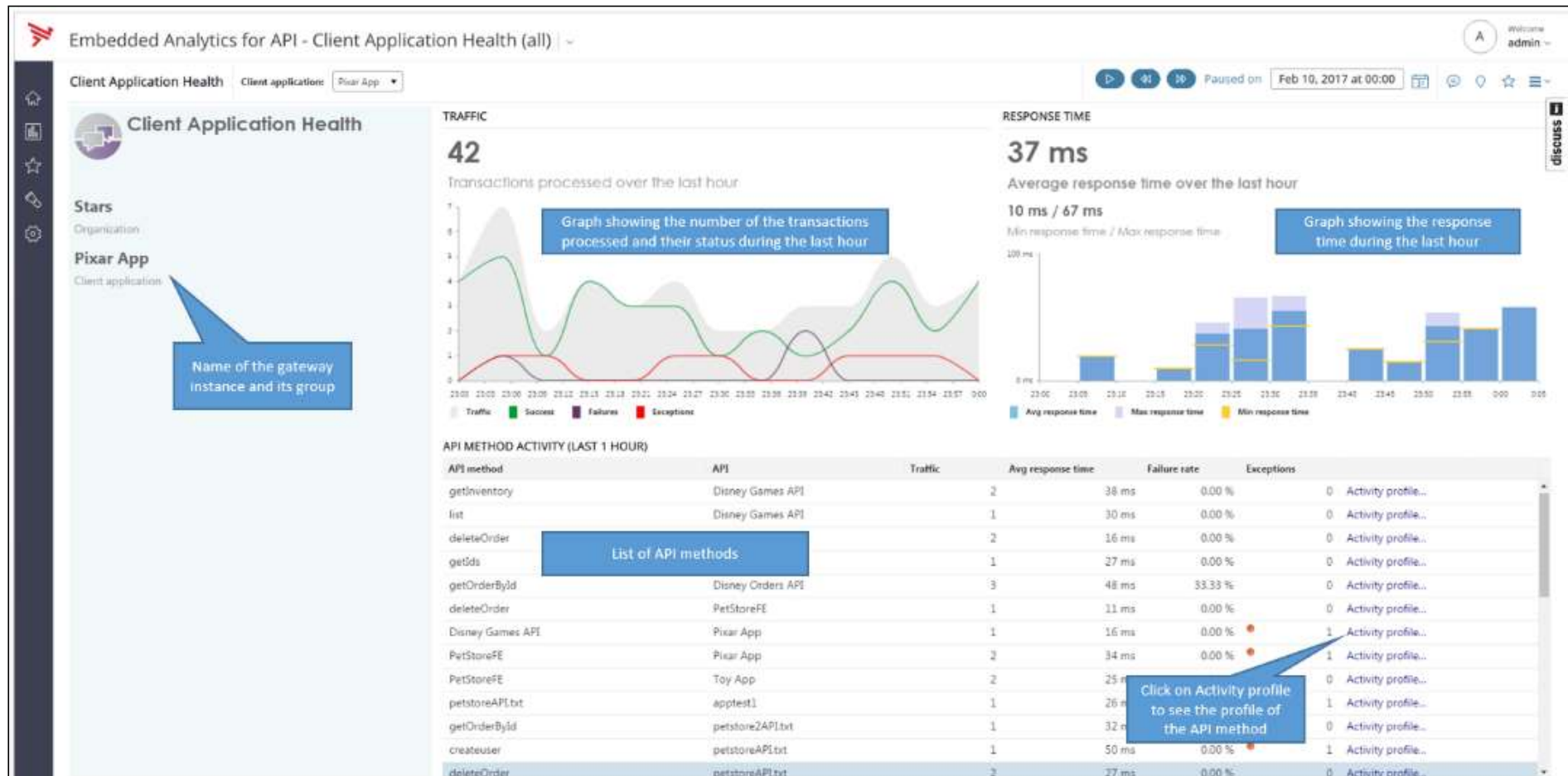
API Infrastructure Health – Gateway instance's health



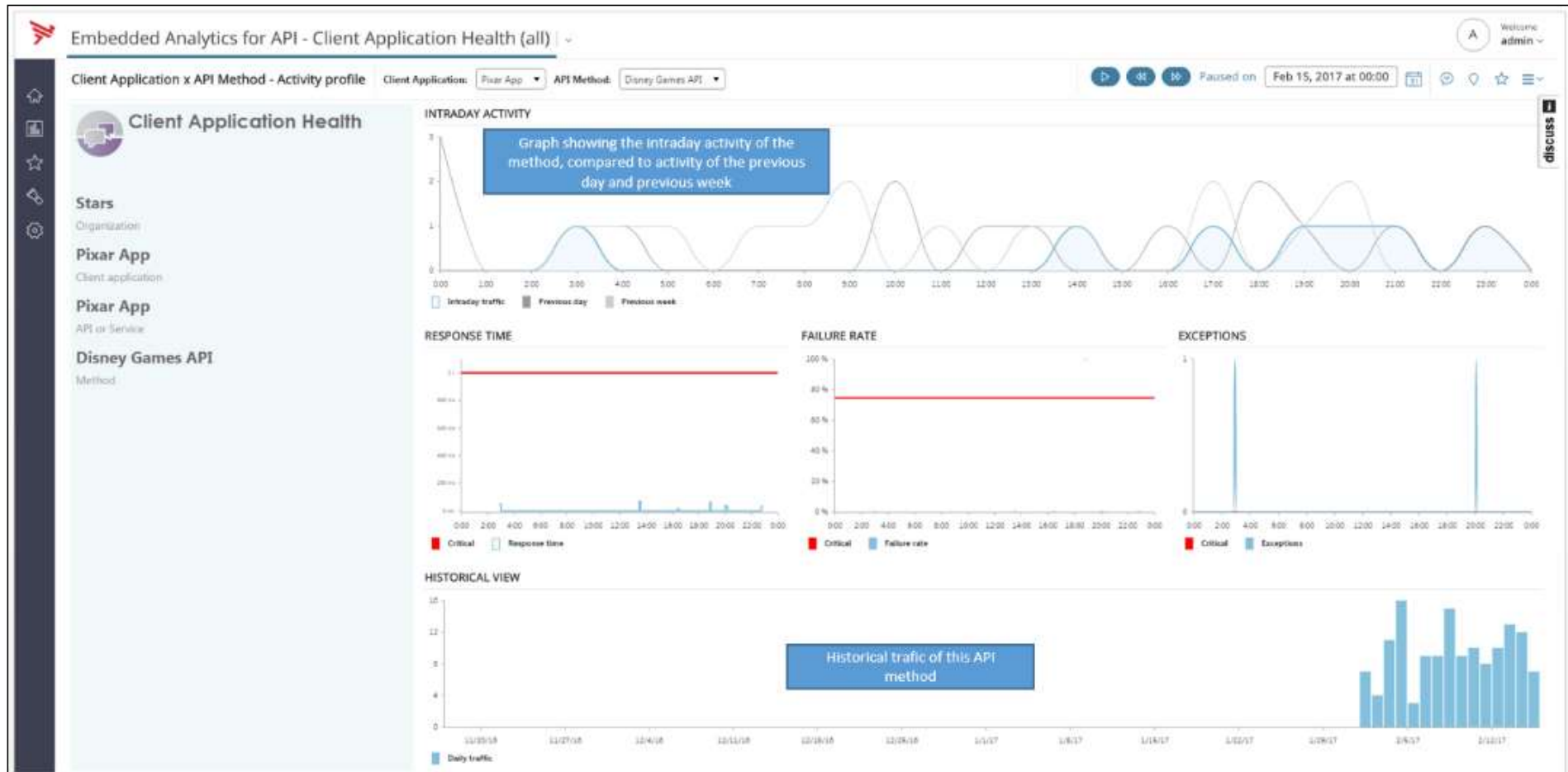
Consumer Engagement - Cockpit



Consumer Engagement – Client application health



Consumer Engagement – API method activity profile



Integration with 3rd party

Integration with 3rd party: for hypervisor

- An IT infrastructure has almost always a hypervisor
- Integrate it, with hypervisor capabilities
 - Process check
 - Port check
- Do not forget regular resources (CPU, memory, disk, ...) checks



Integration with 3rd party: for load-balancer

- Load-balancer is mandatory for HA
- To redirect traffic, load-balancer must know if instance is available
- An XML service is provided in default configuration
- `curl http://<<myhost>>:8080/healthcheck`
- See deployment course for Zero Downtime Deployment configuration

Load Balancer Protocol	Load Balancer Port	Instance Protocol	Instance Port	
HTTP	80	HTTP	80	✕
HTTPS (Secure HTTP)	443	HTTPS (Secure HTTP)	443	✕
Add				



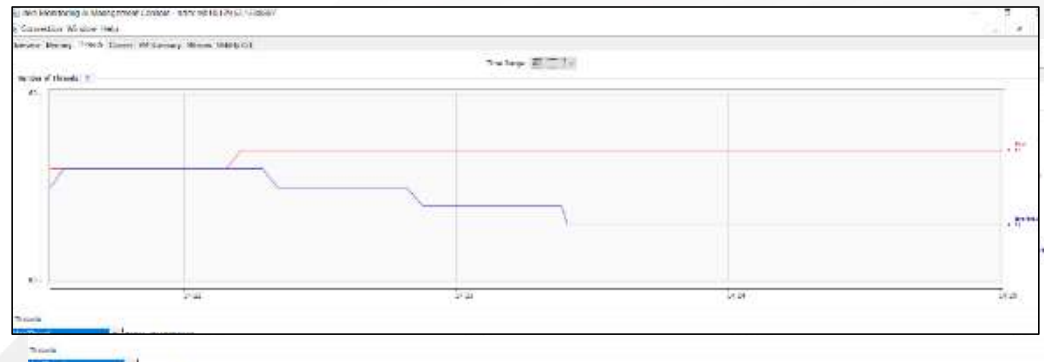
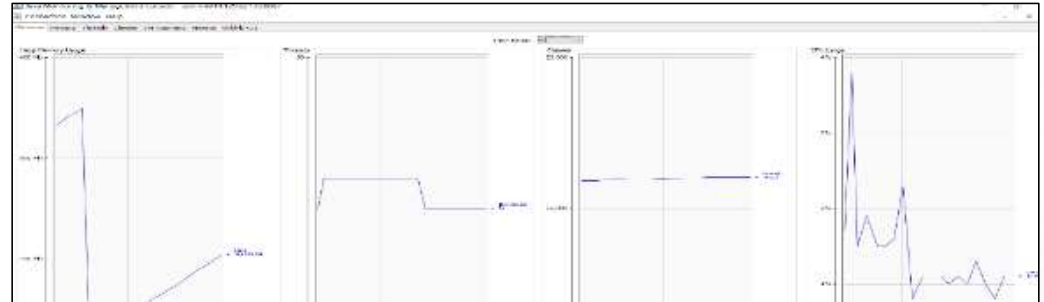
Integration with 3rd party: by APIs

- API Gateway Manager knows instances status
- For more complete integration, other tools can retrieve information
 - See Automation course for more

API Gateway API v1.0		
Contact the developer Copyright Axway 2017. All rights reserved.		
AMA API	Show/Hide	List Operations Expand Operations
Admin Users API	Show/Hide	List Operations Expand Operations
Configuration API	Show/Hide	List Operations Expand Operations
Deployment API	Show/Hide	List Operations Expand Operations
Web Service Discovery	Show/Hide	List Operations Expand Operations
Domain Audit API	Show/Hide	List Operations Expand Operations
Management API	Show/Hide	List Operations Expand Operations
Monitoring API	Show/Hide	List Operations Expand Operations
Role Based Access Control	GET	/router/service/{instance}/api/monitoring/events/{eventType} Gets log, SLA, alert events
Analytics	GET	/router/service/{instance}/api/monitoring/metrics/groups Gets metric groups
Router API	GET	/router/service/{instance}/api/monitoring/metrics/metrics Gets metric information
Topology API	GET	/router/service/{instance}/api/monitoring/metrics/summary Gets metric summary for a Metric Group
KPS API	GET	/router/service/{instance}/api/monitoring/metrics/timeline Gets metric timeline for a Metric Group
	GET	/router/service/{instance}/api/monitoring/metrics/totals Gets metric totals for a Metric Group
	GET	/router/service/{instance}/api/monitoring/summary Gets summary metrics
[BASE URL: /api , API VERSION: 1.0.0]		

Integration with 3rd party: using JMX

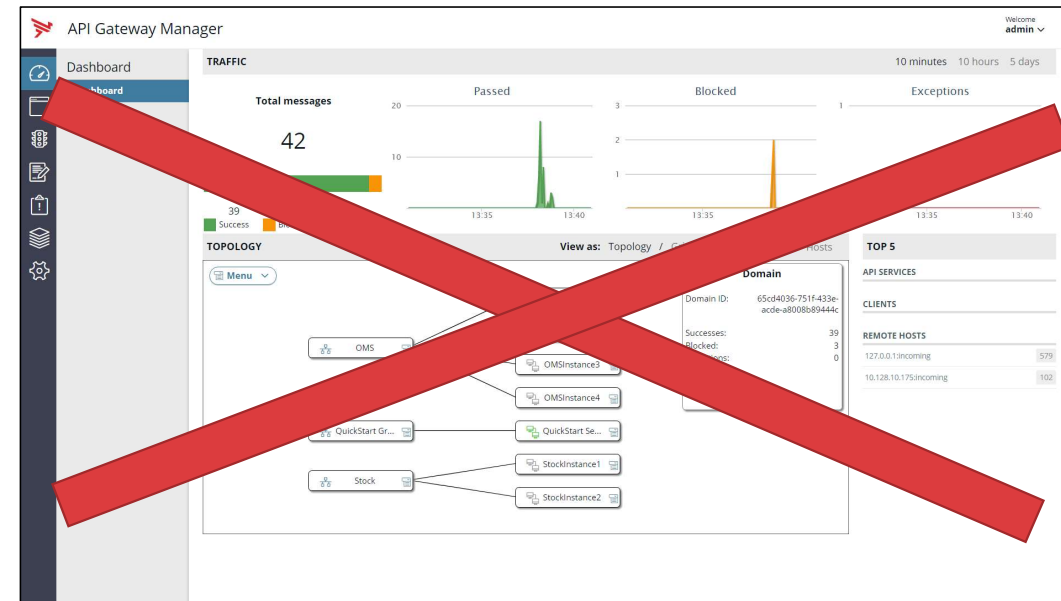
- Advanced method
 - Rarely required
- JMX: Java Management Extensions
 - manage or monitor JVM
- Very precise Java information (thread, objects, garbage collector, ...)
- To be configured in `*/conf/jvm.xml`
 - For NM/ANM
 - For instances



Other cases

Normally, it works...

- As explained, API Gateway Manager is what is needed by an Admin to guarantee traffic is processed correctly
- If not available, try to start it and then use it
- And if not possible, here are some advices



Back to the basics: when you do not know what happens

- Check process: Instance + ANM/NM
- `pgrep vshell`

Process name examples

Node Manager on api-env (Node Manager Group) (7.5.3 SP5) (**vshell**)
QuickStart Server (QuickStart Group) (7.5.3 SP5) (**vshell**)

- Check process: Cassandra
- Process is Java, better check port

- Check port
- `netstat -an |grep 8090`
- Default ports
 - ANM: 8090
 - HTTP default services : 8080
 - API Manager traffic: 8065
 - Cassandra: 9160, 9042, 7000, 7001, 7199
- Note: "-anp" to display pid, but only if processed owned or root

What if... an Admin Node Manager is not running?

- ANM must be HA
 - At least 2 ANM
 - Always one up
- If no ANM is running
 - No more API Gateway Manager
 - No more monitoring
 - No more deployment
 - Traffic processing is working fine
- How to detect?
 - Cannot connect to API Gateway
 - Cannot deploy
- How to check?
 - Process/port
- What to do?
 - Restart it , managedomain -d
- If it fails?
 - Look in apigateway trace

What if... an Instance is not running?

- Instance must be HA
 - At least 2 instances per group
 - Always one up
- If one instance is not running
 - No more access to its monitoring
 - More traffic for other instances
- How to detect?
 - API Gateway Manager, Hypervisor
- How to check?
 - Process/port
- What to do?
 - Restart it, by UI or CLI
- If it fails?
 - Look in trace file, in UI or `apigateway/groups/topologylinks/<Group>-<Instance>/trace`

What if... Cassandra node is not running?

- Cassandra must be HA
 - So 3 nodes minimum
 - Quorum required
 - 3 nodes = minimum 2 running
- If one node is not running
 - It is working, as long the 2 others are up
 - But correct as soon as detected
- How to detect?
 - Hypervisor
- How to check?
 - Process/port
- What to do?
 - See [here](#)

What if... API Portal is not running?

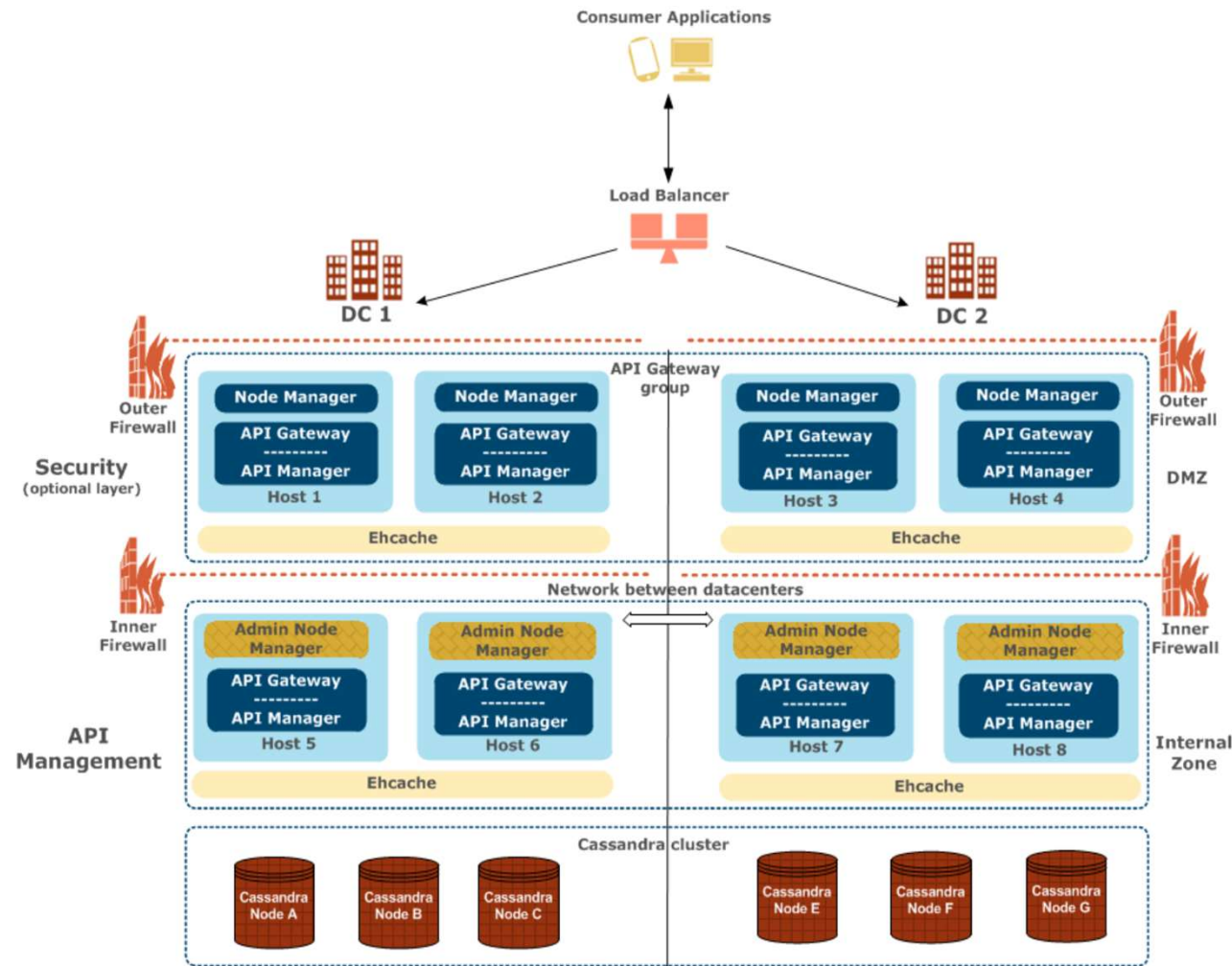
- There can be only one API Portal
 - But if facing outside, be careful about the company image
- If API Portal is not running
 - Users could not access UI
 - No impact on API Catalog and Client Registry
 - No impact on runtime
- How to detect?
 - No connection
 - Hypervisor
- How to check?
 - Process/port
- What to do?
 - Restart it

What if... Datacenter crashed?

- **You'd better have a plan!**
- **(Disaster recovery)**

- API Gateway provides even multi-datacenter clustering
- But disaster recovery requires more than feature, it really requires a plan
- Please, be sure to handle "High Availability" and "Disaster Recovery" consistently, but not as the same topic.

Ex: Arquitetura Multi-Datcenter







Lab – Verificando processos e interfaces





Obrigado